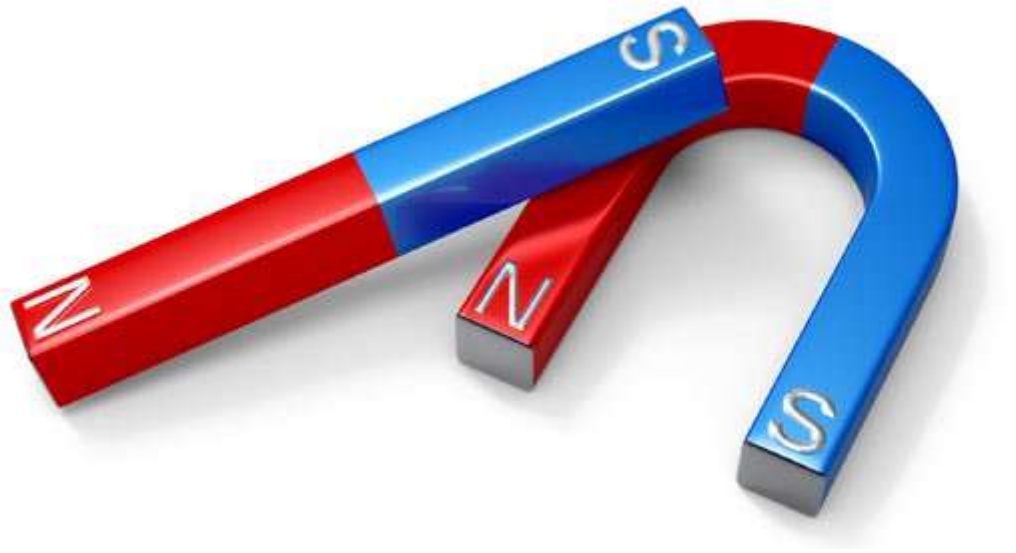


5 Magnetism

In terms of applications, magnetism is one of the most important fields in physics. Large electromagnets are used to pick up heavy loads. Magnets are used in such devices as meters, motors, and loudspeakers. Magnetic tapes and disks are used routinely in sound- and video recording equipment and to store computer data. Intense magnetic fields are used in magnetic resonance imaging (MRI) devices to explore the human body with better resolution and greater safety than x-rays can provide. Giant superconducting magnets are used in the cyclotrons that guide particles into targets at nearly the speed of light. Magnetism is closely linked with electricity. Magnetic fields affect moving charges, and moving charges produce magnetic fields. Changing magnetic fields can even create electric fields. These phenomena signify an underlying unity of electricity and magnetism, which James Clerk Maxwell first described in the 19th century. The ultimate source of any magnetic field is electric current.

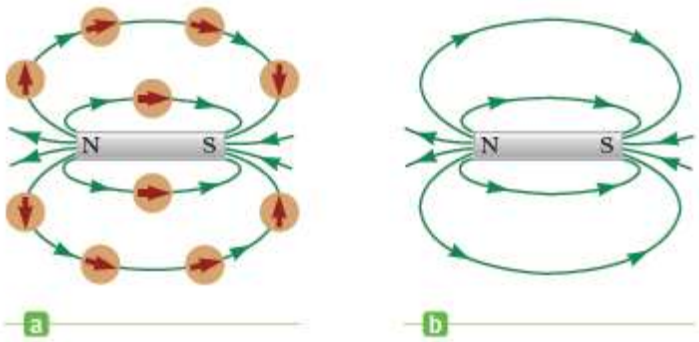
5.1 Magnets

Most people have had experience with some form of magnet. You are most likely familiar with the common iron horseshoe magnet that can pick up iron-containing objects such as paper clips and nails. In the discussion that follows, we assume the magnet has the shape of a bar. Iron objects are most strongly attracted to either end of such a bar magnet, called its **poles**. One end is called the **north pole** and the other the **south pole**. The names come from the behavior of a magnet in the presence of Earth's magnetic field. If a bar magnet is suspended from its midpoint by a piece of string so that it can swing freely in a horizontal plane, it will rotate until its north pole points to the north and its south pole points to the south. The same idea is used to construct a simple compass. Magnetic poles also exert attractive or repulsive forces on each other similar to the electrical forces between charged objects. In fact, simple experiments with two bar magnets show **that like poles repel each other and unlike poles attract each other**.

Although the force between opposite magnetic poles is similar to the force between positive and negative electric charges, there is an important difference: positive and negative electric charges can exist in isolation of each other; north and south poles don't. No matter how many times a permanent magnet is cut, each piece always has a north pole and a south pole. There is some theoretical basis, however, for the speculation that magnetic monopoles (isolated north or south poles) exist in nature, and the attempt to detect them is currently an active experimental field of investigation. An unmagnetized piece of iron can be magnetized by stroking it with a magnet. Magnetism can also be induced in iron (and other materials) by other means. For example, if a piece of unmagnetized iron is placed near a strong permanent magnet, the piece of iron eventually becomes magnetized. The process can be accelerated by heating and then cooling the iron.

Naturally occurring magnetic materials such as magnetite are magnetized in this way because they have been subjected to Earth's magnetic field for long periods of time. The extent to which a piece of material retains its magnetism depends on whether it is classified as magnetically hard or soft. Soft magnetic materials, such as iron, are easily magnetized but tend to lose their magnetization easily. These materials are used in the cores of transformers, generators, and motors.

Iron is the most common choice because it's inexpensive. Other magnetically soft materials include nickel, nickel-iron alloys, and ferrites. Ferrites are combinations of a divalent metal oxide of nickel or magnesium with ferric oxide. Ferrites are used in high-frequency applications, such as radar. Hard magnetic materials are used in permanent magnets. Such magnets provide magnetic fields without the use of electricity. Permanent magnets are used in many devices, including loudspeakers, permanent-magnet motors, and the read/ write heads of computer hard drives. There are a large number of different materials used in permanent magnets. Alnico is a generic name for various alloys of iron, cobalt, and nickel, together with smaller amounts of aluminum, copper, or other elements. Rare earths such as samarium and neodymium are also used in conjunction with other elements to make strong permanent magnets. In earlier chapters we described the interaction between charged objects in terms of electric fields. Recall that an electric field surrounds any stationary electric charge. The region of space surrounding a moving charge includes a magnetic field as well. A magnetic field also surrounds a properly magnetized magnetic material. To describe any type of vector field, we must define its magnitude, or strength, and its direction. The direction of a magnetic field $\vec{B}$at any location is the direction



Active Figure 5.1

in which the north pole of a compass needle points at that location. Active Figure 5.2a shows how the magnetic field of a bar magnet can be traced with the aid of a compass, defining a magnetic field line. Several magnetic field lines of a bar magnet traced out in this way appear in the two-dimensional representation in Active Figure 5.1b. Magnetic field patterns can be displayed by placing small iron filings in the vicinity of a magnet, as in Figure 5.2. Forensic scientists use a technique similar to that shown in Figure 5.2 to find fingerprints at a crime scene. One way to find latent, or invisible, prints is

by sprinkling a powder of iron dust on a surface. The iron adheres to any perspiration or body oils that are present and can be spread around on the surface with a magnetic
brush that never comes into contact with the powder or the surface.

5.2 Earth's Magnetic Field

A small bar magnet is said to have north and south poles, but it's more accurate to say it has a "north-seeking" pole and a "south-seeking" pole. By these expressions, we mean that if such a magnet is used as a compass, one end will "seek," or point to, **the geographic North Pole of Earth and the other end will "seek," or point to, the geographic South Pole of Earth. We conclude that the geographic North Pole of Earth corresponds to a magnetic south pole, and the geographic South Pole of Earth corresponds to a magnetic north pole.**

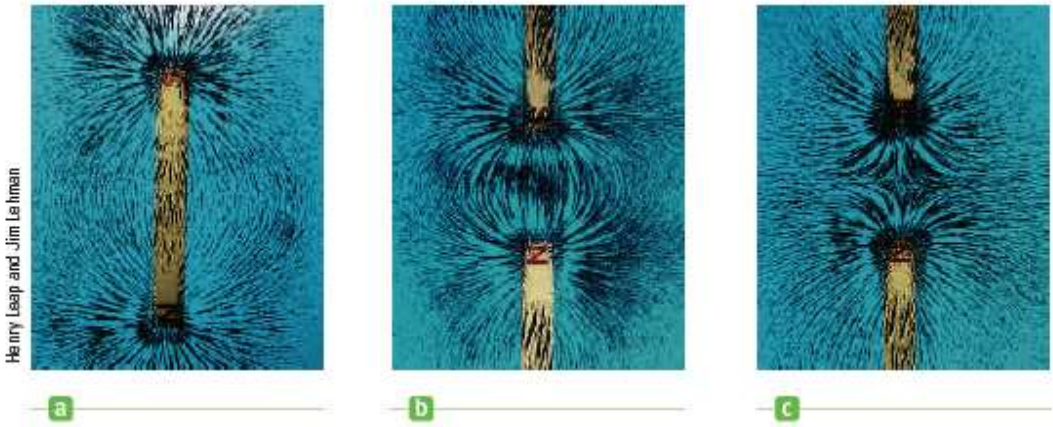


Figure 5.2

In fact, the configuration of Earth's magnetic field, pictured in Figure 5.3, very much resembles what would be observed if a huge bar magnet were buried deep in the Earth's interior. If a compass needle is suspended in bearings that allow it to rotate in the vertical plane as well as in the horizontal plane, the needle is horizontal with respect to Earth's surface only near the equator.

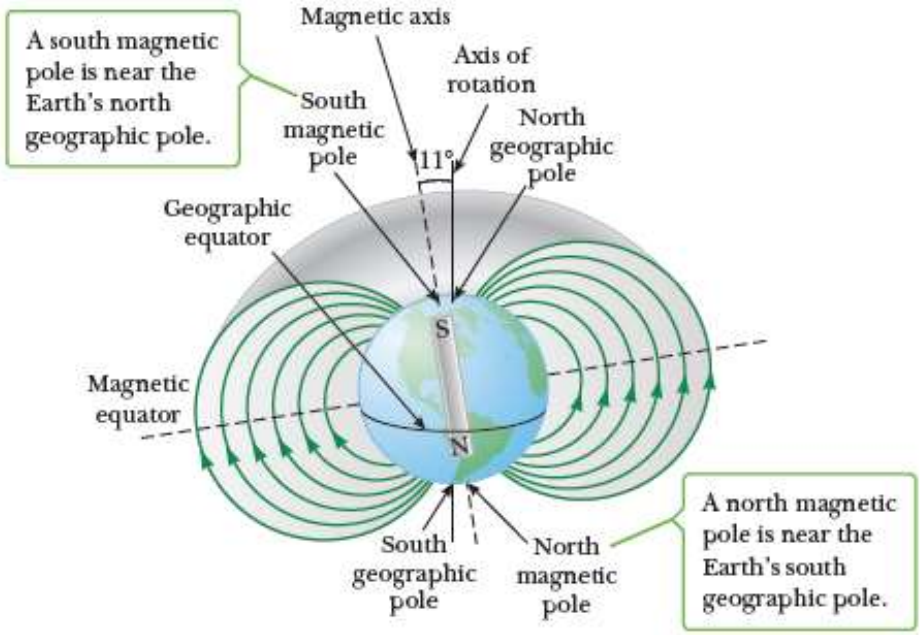


Figure 5.3

As the device is moved northward, the needle rotates so that it points more and more toward the surface of Earth. The angle between the direction of the magnetic field and the horizontal is called the dip angle. Finally, at a point just north of Hudson Bay in Canada, the north pole of the needle points directly downward, with a dip angle of 90°. That site, first found in 1832, is considered to be the location of the south magnetic pole of Earth. It is approximately 1 300 miles from Earth's geographic North Pole and varies with time. Similarly, Earth's magnetic north pole is about 1 200 miles from its geographic South Pole. This means that compass needles point only approximately north. The difference between true north, defined as the geographic North Pole, and north indicated by a compass varies from point to point on Earth, a difference referred to as magnetic declination. For example, along a line through South Carolina and the Great Lakes a compass indicates true north, whereas in Washington state it aligns 25° east of true north (Fig. 5.4).



Figure 5.4

Although the magnetic field pattern of Earth is similar to the pattern that would be set up by a bar magnet placed at its center, the source of Earth's field can't consist of large masses of permanently magnetized material. Earth does have large deposits of iron ore deep beneath its surface, but the high temperatures in the core prevent the iron from retaining any permanent magnetization. It's considered more likely that the true source of Earth's magnetic field is electric current in the liquid part of its core. This current, which is not well understood, may be driven by an interaction between the planet's rotation and convection in the hot liquid core. There is some evidence that the strength of a planet's magnetic field is related to the planet's rate of rotation. For example, Jupiter rotates faster than Earth, and recent space probes indicate that Jupiter's magnetic field is stronger than Earth's, even though Jupiter lacks an iron core. Venus, on the other hand, rotates more slowly than Earth, and its magnetic field is weaker. Investigation into the cause of Earth's magnetism continues.

An interesting fact concerning Earth's magnetic field is that its direction reverses every few million years. Evidence for this phenomenon is provided by basalt (an iron-containing rock) that is sometimes spewed forth by volcanic activity on the ocean floor. As the lava cools, it solidifies and retains a picture of the direction of Earth's magnetic field. When the basalt deposits are dated, they provide evidence for periodic reversals of the magnetic field. The cause of these field reversals is still not understood. It has long been speculated that some animals, such as birds, use the magnetic field of Earth to guide their migrations. Studies have shown that a type of anaerobic bacterium that lives in swamps has a magnetized chain of magnetite as part of its internal structure. (The term anaerobic means that these bacteria live and grow without oxygen; in fact, oxygen is toxic to them.) The magnetized chain acts as a compass needle that enables the bacteria to align with Earth's magnetic field. When they find themselves out of the mud on the bottom of the swamp, they return to their oxygen-free environment by following the magnetic field lines of Earth. Further evidence for their magnetic sensing ability is that bacteria found in the northern hemisphere have internal magnetite chains that are opposite in polarity to those of similar bacteria in the southern hemisphere. Similarly, in the northern hemisphere, Earth's field has a downward component, whereas in the southern hemisphere it has an upward component. Recently, a meteorite originating on Mars has been found

to contain a chain of magnetite. NASA scientists believe it may be a fossil of ancient Martian bacterial life. The magnetic field of Earth is used to label runways at airports according to their direction. A large number is painted on the end of the runway so that it can be read by the pilot of an incoming airplane. This number describes the direction in which the airplane is traveling, expressed as the magnetic heading, in degrees measured clockwise from magnetic north divided by 10. A runway marked 9 would be directed toward the east (90° divided by 10), whereas a runway marked 18 would be directed toward magnetic south.

5.3 Magnetic Field

Experiments show that a stationary charged particle doesn't interact with a static magnetic field. **When a charged particle is moving through a magnetic field, however, a magnetic force acts on it**. This force has its maximum value when the charge moves in a direction perpendicular to the magnetic field lines, decreases in value at other angles, and becomes zero when the particle moves along the field lines. This is quite different from the electric force, which exerts a force on a charged particle whether it's moving or at rest. Further, the electric force is directed parallel to the electric field whereas the magnetic force on a moving charge is directed perpendicular to the magnetic field.

In our discussion of electricity, the electric field at some point in space was defined as the electric force per unit charge acting on some test charge placed at that point. In a similar way, we can describe the properties of the magnetic field $\vec{B}$ at some point in terms of the magnetic force exerted on a test charge at that point. Our test object is a charge q moving with velocity $\vec{v}$. It is found experimentally that the strength of the magnetic force on the particle is proportional to the magnitude of the charge q, the magnitude of the velocity $\vec{v}$, the strength of the external magnetic field $\vec{B}$, and the sine of the angleθ between the direction of $\vec{v}$ and the direction of $\vec{B}$. These observations can be summarized by writing the magnitude of the magnetic force as

$$F = qvB\sin\theta \qquad [5.1]$$

This expression is used to define the magnitude of the magnetic field as

$$B \equiv \frac{F}{qv\sin\theta} \qquad [5.2]$$

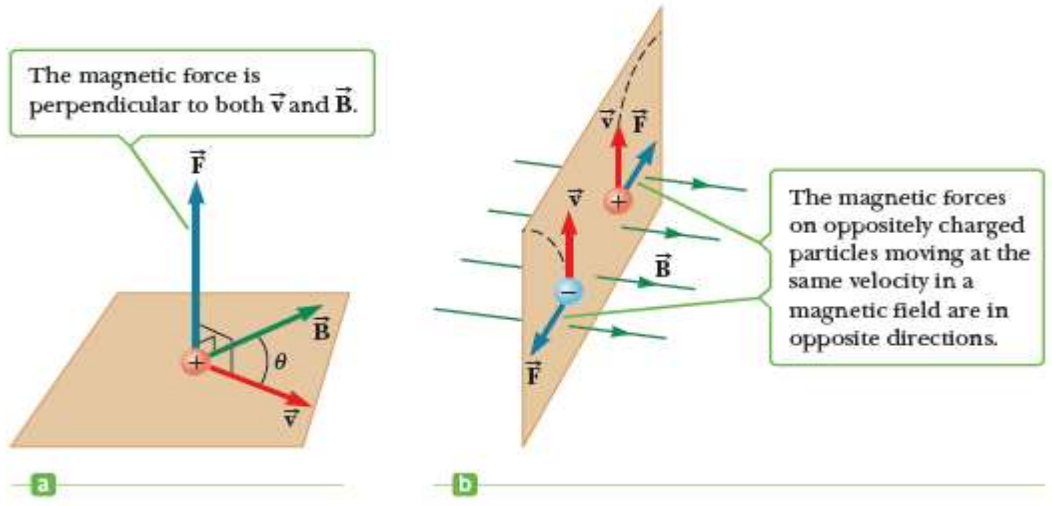


Figure 5.5 (a) The direction of the magnetic force $\vec{F}$ acting on a charged particle moving with a velocity $\vec{V}$ in the presence of a magnetic field $\vec{B}$. (b) Magnetic forces on positive and negative charges. The dashed lines show the paths of the particles

If F is in newtons, q in coulombs, and v in meters per second, the SI unit of magnetic field is the **tesla (T),** also called the **weber (Wb) per square meter** (1 T = 1 Wb/m2). If a 1-C charge moves in a direction perpendicular to a magnetic field of magnitude 1 T with a speed of 1 m/s, the magnetic force exerted on the charge is 1 N. We can express the units of $\vec{B}$ as

$$[\mathrm{B}] = \mathrm{T} = \frac{\mathrm{Wb}}{\mathrm{m}^2} = \frac{\mathrm{N}}{\mathrm{C} \cdot \mathrm{m/s}} = \frac{\mathrm{N}}{\mathrm{A} \cdot \mathrm{m}} \qquad [5.3]$$

In practice, the cgs unit for magnetic field, the **gauss** (G), is often used. The gauss is related to the tesla through the conversion

$$1\ \mathrm{T} = 10^4\ \mathrm{G}$$

Conventional laboratory magnets can produce magnetic fields as large as about 25 000 G, or 2.5 T. Superconducting magnets that can generate magnetic fields as great as 3×10^5 G, or 30 T, have been constructed. These values can be compared with the small value of Earth's magnetic field near its surface, which is only about 0.5 G, or 0.5×10^{-4} T.

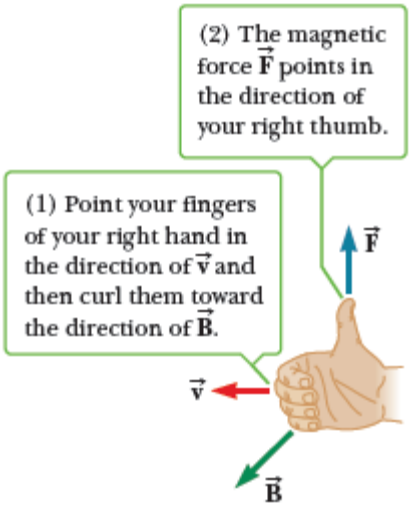
From Equation 5.1 we see that the force on a charged particle moving in a magnetic field has its maximum value when the particle's motion is perpendicular to the magnetic field, corresponding to $\theta = 90°$, so that $\sin\theta = 1$. The magnitude of this maximum force has the value

$$F_{\max} = qvB \qquad [5.4]$$

Also from Equation 5.1, F is zero when $\vec{V}$ is parallel to $\vec{B}$ (corresponding to $\theta = 0°$ or 180°), so no magnetic force is exerted on a charged particle when it moves in the direction of the magnetic field or opposite the field.

Experiments show that the direction of the magnetic force is always perpendicular to both $\vec{V}$ and $\vec{B}$, as shown in Figure 5.5 for a positively charged particle. To determine the direction of the force, we employ **right-hand rule number 1**:

1. *Point the fingers of your right hand in the direction of the velocity* $\vec{V}$
2. *Curl the fingers in the direction of the magnetic field* $\vec{B}$*, moving through the smallest angle (as in Fig. 5.6).*
3. *Your thumb is now pointing in the direction of the magnetic force* $\vec{F}$ *exerted on a positive charge.*



If the charge is negative rather than positive, the force $\vec{F}$ is directed opposite that shown in Figures 5.5a and 5.6. So if q is negative, simply use the right-hand rule to find the direction for positive q and then reverse that direction for the negative charge. Figure 5.5b illustrates the effect of a magnetic field on charged particles with opposite signs.

PROBLEM A proton moves with a speed of 1.00×10^5 m/s through Earth's magnetic field, which has a value of 55.0 μT at a particular location. When the proton moves eastward, the magnetic force acting on it is directed straight upward, and when it moves northward, no magnetic force acts on it. (a) What is the direction of the magnetic field, and (b) what is the strength of the magnetic force when the proton moves eastward? (c) Calculate the gravitational force on the proton and compare it with the magnetic force. Compare it also with the electric force if there were an electric field with a magnitude equal to $E = 1.50 \times 10^2$ N/C at that location, a common value at Earth's surface. Note that the mass of the proton is 1.67×10^{-27} kg.

STRATEGY The direction of the magnetic field can be found from an application of the right-hand rule, together with the fact that no force is exerted on the proton when it's traveling north. Substituting into Equation 5.1 yields the magnitude of the magnetic field.

SOLUTION

(a) Find the direction of the magnetic field.

No magnetic force acts on the proton when it's going north, so the angle such a proton makes with the magnetic field direction must be either 0° or 180°. Therefore, the magnetic field $\vec{\mathbf{B}}$ must point either north or south. Now apply the right-hand rule. When the particle travels east, the magnetic force is directed upward. Point your thumb in the direction of the force and your fingers in the direction of the velocity eastward. When you curl your fingers, they point north, which must therefore be the direction of the magnetic field.

(b) Find the magnitude of the magnetic force.

Substitute the given values and the charge of a proton into Equation 19.1. From part (a), the angle between the velocity $\vec{\mathbf{v}}$ of the proton and the magnetic field $\vec{\mathbf{B}}$ is 90.0°.

$$F = qvB \sin\theta$$
$$= (1.60 \times 10^{-19}\ \text{C})(1.00 \times 10^5\ \text{m/s}) \times (55.0 \times 10^{-6}\ \text{T}) \sin(90.0°)$$
$$= \boxed{8.80 \times 10^{-19}\ \text{N}}$$

(c) Calculate the gravitational force on the proton and compare it with the magnetic force and also with the electric force if $E = 1.50 \times 10^2$ N/C.

$$F_{\text{grav}} = mg = (1.67 \times 10^{-27}\ \text{kg})(9.80\ \text{m/s}^2)$$
$$= \boxed{1.64 \times 10^{-26}\ \text{N}}$$
$$F_{\text{elec}} = qE = (1.60 \times 10^{-19}\ \text{C})(1.50 \times 10^2\ \text{N/C})$$
$$= \boxed{2.40 \times 10^{-17}\ \text{N}}$$

REMARKS The information regarding a proton moving north was necessary to fix the direction of the magnetic field. Otherwise, an upward magnetic force on an eastward-moving proton could be caused by a magnetic field pointing anywhere northeast or northwest. Notice in part (c) the relative strengths of the forces, with the electric force larger than the magnetic force and both much larger than the gravitational force, all for typical field values found in nature.

5.4 Magnetic Force on a Current-Carrying Conductor

If a magnetic field exerts a force on a single charged particle when it moves through a magnetic field, it should be no surprise that magnetic forces are exerted on a current-carrying wire as well (see Fig. 5.9). Because the current is a collection of many charged particles in motion, the resultant force on the wire is due to the sum of the individual forces on the charged particles. The force on the particles is transmitted to the "bulk" of the wire through collisions with the atoms making up the wire. Some explanation is in order concerning notation in many of the figures. To indicate the direction of $\vec{B}$ we use the following conventions:

If $\vec{B}$ is directed into the page, as in Figure 5.10, we use a series of green crosses, representing the tails of arrows. If $\vec{B}$ is directed out of the page, we use a series of green dots, representing the tips of arrows. If $\vec{B}$ lies in the plane of the page, we use a series of green field lines with arrowheads.

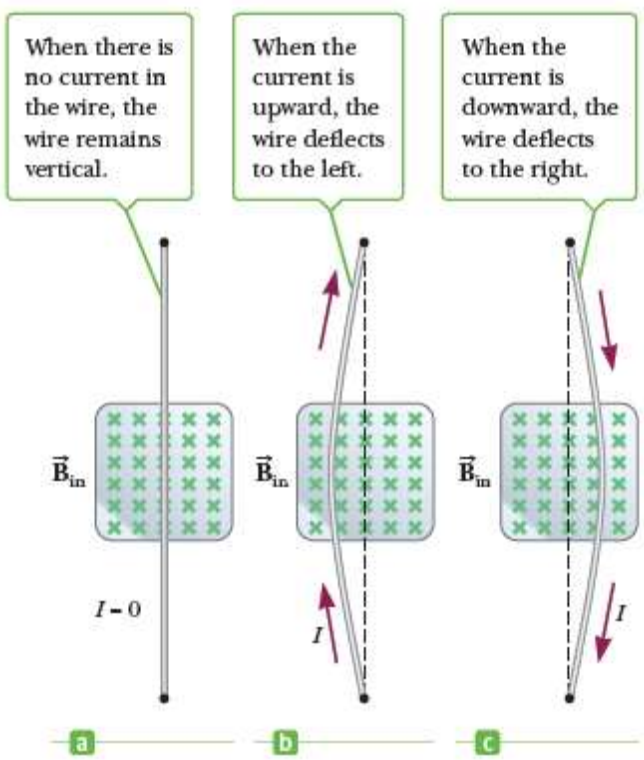
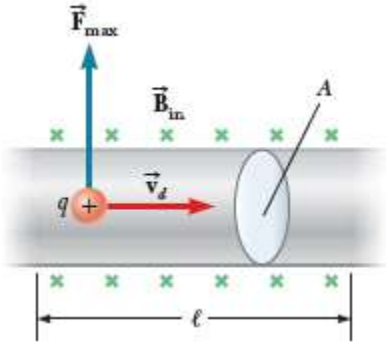


Figure 5.10 A segment of a flexible vertical wire partially stretched between the poles of a magnet,with the field (green crosses) directed into the page.

The force on a current-carrying conductor can be demonstrated by hanging a wire between the poles of a magnet, as in Figure 5.10. In this figure, the magnetic field is directed into the page and covers the region within the shaded area. The wire deflects to the right or left when it carries a current.
We can quantify this discussion by considering a straight segment of wire of length , and cross-sectional area A carrying current I in a uniform external magnetic field $\vec{B}$, as in Figure 5.11. We assume that the magnetic field is perpendicular to the wire and is directed into the page. A force of magnitude $F_{max} = qv_dB$ is exerted on each charge carrier in the wire, where vd is the drift velocity of the charge. To find the total force on the wire, we multiply the force on one charge carrier by the number of carriers in the segment. Because the volume of the segment is Al the number of carriers is nAl, where n is the number of carriers per unit volume. Hence, the magnitude of the total magnetic force on the wire of length l is as follows:

Total force = force on each charge carrier × total number of carriers



$$F_{max} = (qv_dB)(nA\ell)$$

however, we know that the current in the wire is given by the expression $I = nqv_dA,$

$$F_{max} = BI\ell \qquad [5.5]$$

This equation can be used only when the current and the magnetic field are at right angles to each other.

Figure 5.11 A section of a wire containing moving charges in an external magnetic field $\vec{B}$
.

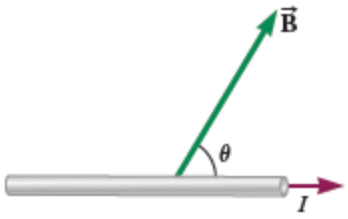


Figure 19.12 A wire carrying a current I in the presence of an external magnetic field $\vec{B}$ that makes an angle u with the wire. The magnetic force vector comes out of the page.

If the wire is not perpendicular to the field but is at some arbitrary angle, as in Figure 5.12, the magnitude of the magnetic force on the wire is

$$F = BI\ell \sin\theta \qquad [5.6]$$

where u is the angle between $\vec{B}$ and the direction of the current. The direction of this force can be obtained by the use of right-hand rule number 1. In this case, however, you must place your fingers in the direction of the positive current I, rather than in the direction of $\vec{V}$, before curling them in the direction of $\vec{B}$. The thumb then points in the direction of the force, as before. The current, naturally, is made up of charges moving at some velocity, so this really isn't a separate rule. In Figure 5.12 the direction of the magnetic force on the wire is out of the page.

Finally, when the current is either in the direction of the field or opposite the direction of the field, the magnetic force on the wire is zero. A magnetic force acting on a current-carrying wire in a magnetic field is the operating principle of most speakers in sound systems. One speaker design, shown in Figure 5.13, consists of a coil of wire called the voice coil, a flexible paper cone that acts as the speaker, and a permanent magnet. The coil of wire surrounding the north pole of the magnet is shaped so that the magnetic field lines are directed radially outward from the coil's axis. When an electrical signal is sent to the coil, producing a current in the coil as in Figure 5.13, a magnetic force to the left acts on the coil. (This can be seen by applying right-hand rule number 1 to each turn of wire.) When the current reverses direction, as it would for a current that varied sinusoidally, the magnetic force on the coil also reverses direction, and the cone, which is attached to the coil, accelerates to the right. An alternating current through the coil causes an alternating force on the coil, which results in vibrations of the cone. The vibrating cone creates sound waves as it pushes and pulls on the air in front of it. In this way, a 1-kHz electrical signal is converted to a 1-kHz sound wave.

An application of the force on a current-carrying conductor is illustrated by the electromagnetic pump shown in Figure 5.14. Artificial hearts require a pump to keep the blood flowing, and kidney dialysis machines also require a pump to assist the heart in pumping blood that is to be cleansed. Ordinary mechanical pumps create problems because they damage the blood cells as they move through the pump. The mechanism shown in the figure has demonstrated some promise in such applications. A magnetic field is established across a segment of the tube containing the blood, flowing in the direction of the velocity $\vec{V}$. An electric current passing through the fluid in the direction shown has a magnetic force acting on it in the direction of $\vec{V}$, as applying the right-hand rule shows. This force helps to keep the blood in motion.

PROBLEM A wire carries a current of 22.0 A from west to east. Assume the magnetic field of Earth at this location is horizontal and directed from south to north and it has a magnitude of 0.500×10^{-4} T. (a) Find the magnitude and direction of the magnetic force on a 36.0-m length of wire. (b) Calculate the gravitational force on the same length of wire if it's made of copper and has a cross-sectional area of 2.50×10^{-6} m^2.

SOLUTION

(a) Calculate the magnetic force on the wire.

Substitute into Equation 19.6, using the fact that the magnetic field and the current are at right angles to each other:

$$F = BI\ell \sin\theta = (0.500 \times 10^{-4}\ \text{T})(22.0\ \text{A})(36.0\ \text{m}) \sin 90.0^\circ$$
$$= 3.96 \times 10^{-2}\ \text{N}$$

Apply right-hand rule number 1 to find the direction of the magnetic force:

With the fingers of your right hand pointing west to east in the direction of the current, curl them north in the direction of the magnetic field. Your thumb points upward.

(b) Calculate the gravitational force on the wire segment.

First, obtain the mass of the wire from the density of copper, the length, and cross-sectional area of the wire:

$$m = \rho V = \rho(A\ell)$$
$$= (8.92 \times 10^3\ \text{kg/m}^3)(2.50 \times 10^{-6}\ \text{m}^2 \cdot 36.0\ \text{m})$$
$$= 0.803\ \text{kg}$$

To get the gravitational force, multiply the mass by the acceleration of gravity:

$$F_{\text{grav}} = mg = 7.87\ \text{N}$$

REMARKS This calculation demonstrates that under normal circumstances, the gravitational force on a current-carrying conductor is much greater than the magnetic force due to Earth's magnetic field.

5.5 Torque on a Current Loop and Electric Motors

In the preceding section we showed how a magnetic force is exerted on a current carrying conductor when the conductor is placed in an external magnetic field. With this starting point, we now show that a torque is exerted on a current loop placed in a magnetic field. The results of this analysis will be of great practical value when we discuss generators and motors in Chapter 20.

Consider a rectangular loop carrying current I in the presence of an external uniform magnetic field in the plane of the loop, as shown in Figure 5.15a. The forces on the sides of length a are zero because these wires are parallel to the field. The magnitudes of the magnetic forces on the sides of length b, however, are

$$F_1 = F_2 = BIb$$

The direction of $\overrightarrow{F1}$, the force on the left side of the loop, is out of the page and that of $\overrightarrow{F2}$, the force on the right side of the loop, is into the page. If we view the loop from the side, as in Figure 5.15b, the forces are directed as shown. If we

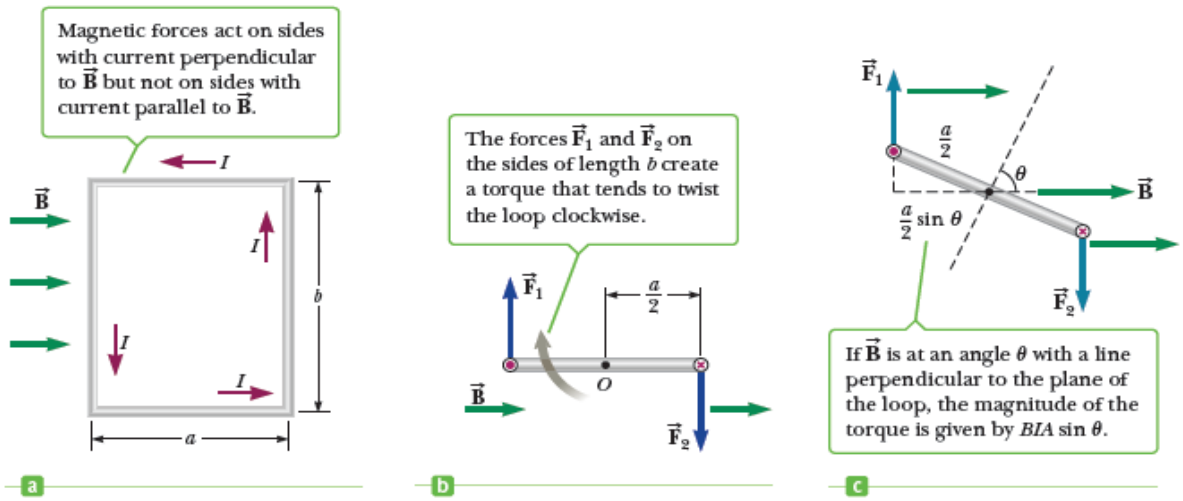


Figure 5.15

assume the loop is pivoted so that it can rotate about point O, we see that these two forces produce a torque about O that rotates the loop clockwise. The magnitude of this torque, τ_{max}, is

$$\tau_{\max} = F_1 \frac{a}{2} + F_2 \frac{a}{2} = (BIb)\frac{a}{2} + (BIb)\frac{a}{2} = BIab$$

where the moment arm about O is a/2 for both forces. Because the area of the loop is A = ab, the maximum torque can be expressed as

$$\tau_{\max} = BIA \qquad [5.7]$$

This result is valid only when the magnetic field is parallel to the plane of the loop, as in Figure 5.15b. If the field makes an angle u with a line perpendicular to the plane of the loop, as in Figure 5.15c, the moment arm for each force is given by (a/2) sinθ. An analysis similar to the previous one gives, for the magnitude of the torque,

$$\tau = BIA \sin\theta \qquad [5.8]$$

This result shows that the torque has the maximum value BIA when the field is parallel to the plane of the loop (θ = 90°) and is zero when the field is perpendicular to the plane of the loop (θ = 0). As seen in Figure 5.15c, the loop tends to rotate to smaller values of u (so that the normal to the plane of the loop rotates toward the direction of the magnetic field).

Although the foregoing analysis was for a rectangular loop, a more general derivation shows that Equation 5.8 applies regardless of the shape of the loop. Further, the torque on a coil with N turns is

$$\tau = BIAN\sin\theta \qquad [5.9a]$$

The quantity μ = IAN is defined as the magnitude of a vector $\vec{\mu}$ called the magnetic moment of the coil. The vector $\vec{\mu}$ always points perpendicular to the plane of the loop(s) and is such that if the thumb of the right hand points in the direction of $\vec{\mu}$, the fingers of the right hand point in the direction of the current. The angle θ in Equations 5.8 and 5.9 lies between the directions of the magnetic moment $\vec{\mu}$ and the magnetic field $\vec{B}$. The magnetic torque can then be written

$$\tau = \mu B\sin\theta \qquad [5.9b]$$

Note that the torque $\vec{\tau}$ is always perpendicular to both the magnetic moment $\vec{\mu}$ and the magnetic field $\vec{B}$.

PROBLEM A circular wire loop of radius 1.00 m is placed in a magnetic field of magnitude 0.500 T. The normal to the plane of the loop makes an angle of 30.0° with the magnetic field (Fig. 5.16a). The current in the loop is 2.00 A in the direction shown. (a) Find the magnetic moment of the loop and the magnitude of the torque at this instant. (b) The same current is carried by the rectangular 2.00-m by 3.00-m coil with three loops shown in Figure 5.16b. Find the magnetic moment of the coil and the magnitude of the torque acting on the coil at that instant.

STRATEGY For each part, we just have to calculate the area, use it in the calculation of the magnetic moment, and multiply the result by Bsinθ. Altogether, this process amounts to substituting values into Equation 5.9b.

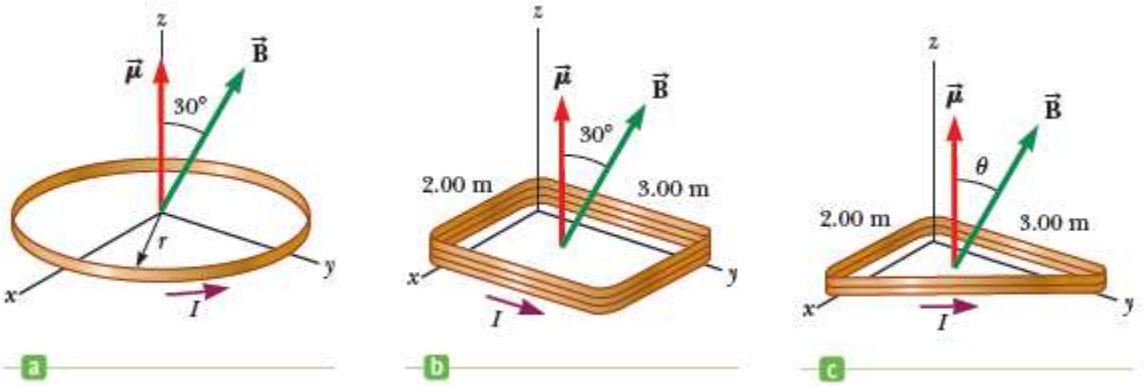


Figure 5.16 (a) A circular current loop lying in the xy-plane in an external magnetic field $\vec{B}$. (b) A rectangular coil lying in the xy-plane in the same field.

SOLUTION

(a) Find the magnetic moment of the circular loop and the magnetic torque exerted on it.

First, calculate the enclosed area of the circular loop: $A = \pi r^2 = \pi(1.00\ \text{m})^2 = 3.14\ \text{m}^2$

Calculate the magnetic moment of the loop: $\mu = IAN = (2.00\ \text{A})(3.14\ \text{m}^2)(1) = \boxed{6.28\ \text{A} \cdot \text{m}^2}$

Now substitute values for the magnetic moment, magnetic field, and θ into Equation 19.9b: $\tau = \mu B \sin\theta = (6.28\ \text{A} \cdot \text{m}^2)(0.500\ \text{T})(\sin 30.0°) = \boxed{1.57\ \text{N} \cdot \text{m}}$

(b) Find the magnetic moment of the rectangular coil and the magnetic torque exerted on it.

Calculate the area of the coil: $A = L \times H = (2.00\ \text{m})(3.00\ \text{m}) = 6.00\ \text{m}^2$

Calculate the magnetic moment of the coil: $\mu = IAN = (2.00\ \text{A})(6.00\ \text{m}^2)(3) = \boxed{36.0\ \text{A} \cdot \text{m}^2)}$

Substitute values into Equation 19.9b: $\tau = \mu B \sin\theta = (0.500\ \text{T})(36.0\ \text{A} \cdot \text{m}^2)(\sin 30.0°) = \boxed{9.00\ \text{N} \cdot \text{m}}$

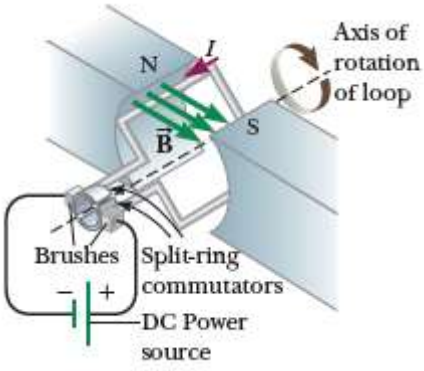
Electric Motors

It's hard to imagine life in the 21st century without electric motors. Some appliances that contain motors include computer disk drives, CD players, DVD players, food processors and blenders, car starters, furnaces, and air conditioners. The motors convert electrical energy to kinetic energy of rotation and consist of a rigid current-carrying loop that rotates when placed in the field of a magnet.

As we have just seen (Fig. 5.15), the torque on such a loop rotates the loop to smaller values of u until the torque becomes zero, when the magnetic field is perpendicular to the plane of the loop and $\theta = 0$. If the loop turns past this angle and the current remains in the direction shown in the figure, the torque reverses direction and turns the loop in the opposite direction, that is, counterclockwise. To overcome this difficulty and provide continuous rotation in one direction, the current in the loop must periodically reverse direction. In alternating-current (AC) motors, such a reversal occurs naturally 120 times each second. In direct current (DC) motors, the reversal is accomplished mechanically with split-ring contacts (commutators) and brushes, as shown in Active Figure 5.17.

Although actual motors contain many current loops and commutators, for simplicity Active Figure 5.17 shows only a single loop and a single set of split-ring contacts rigidly attached to and rotating with the loop. Electrical stationary contacts called brushes are maintained in electrical contact with the rotating split ring. These brushes are usually made of graphite because it is a good electrical

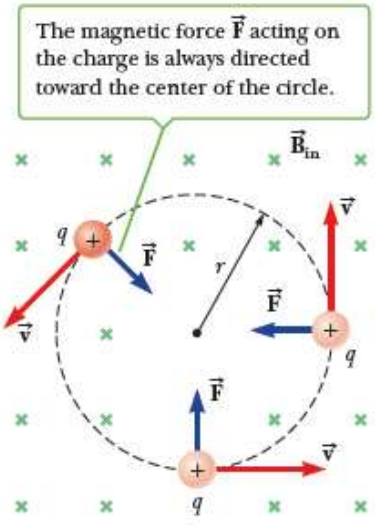
conductor as well as a good lubricant. Just as the loop becomes perpendicular to the magnetic field and the torque becomes zero, inertia carries the loop forward in the clockwise direction and the brushes cross the gaps in the ring, causing the loop current to reverse its direction. This reversal provides another pulse of torque in the clockwise direction for another 180°, the current reverses, and the process repeats itself. Figure 5.18 shows a modern motor used to power a hybrid gas–electric car.



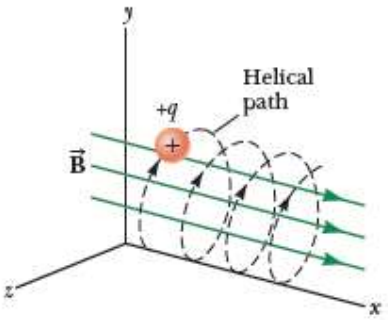
Active Figure 4.17 Simplified sketch of a DC electric motor.

5.6 Motion of a Charged Particle in a Magnetic Field

Consider the case of a positively charged particle moving in a uniform magnetic field so that the direction of the particle's velocity is perpendicular to the field, as in Active Figure 5.19. The label $\overrightarrow{B_{in}}$ in and the crosses indicate that $\vec{B}$ is directed into the page. Application of the right-hand rule to the particle at the bottom of the circle shows that the direction of the magnetic force $\vec{F}$ at that location is upward. The force causes the particle to alter its direction of travel and to follow a curved path. Application of the right-hand rule to the particle at other points on the circle shows that the magnetic force is always directed toward the center of the circular path; therefore, the magnetic force causes a centripetal acceleration, which changes only the direction of $\vec{V}$ and not its magnitude. Because $\vec{F}$ produces the centripetal acceleration, we can equate its magnitude, qvB in this case, to the mass of the particle multiplied by the centripetal acceleration v^2/r. From Newton's second law, we find that



Active Figure 5.18



Active Figure 5.19

$$F = qvB = \frac{mv^2}{r}$$ which gives $$r = \frac{mv}{qB}$$ [5.10]

This equation says that the radius of the path is proportional to the momentum mv of the particle and is inversely proportional to the charge and the magnetic field. Equation 5.10 is often called the cyclotron equation because it's used in the design of these instruments (popularly known as atom smashers).

If the initial direction of the velocity of the charged particle is not perpendicular to the magnetic field, as shown in Active Figure 5.20, the path followed by the particle is a spiral (called a helix) along the magnetic field lines.

PROBLEM A charged particle enters the magnetic field of a mass spectrometer at a speed of 1.79×10^6 m/s. It subsequently moves in a circular orbit with a radius of 16.0 cm in a uniform magnetic field of magnitude 0.350 T having a direction perpendicular to the particle's velocity. Find the particle's mass-to-charge ratio and identify it based on the table below.

STRATEGY After finding the mass-to-charge ratio with Equation 5.10, compare it with the values in the table, identifying the particle.

SOLUTION

Write the cyclotron equation: $$r = \frac{mv}{qB}$$

Solve this equation for the mass divided by the charge, m/q, and substitute values: $$\frac{m}{q} = \frac{rB}{v} = \frac{(0.160\ \text{m})(0.350\ \text{T})}{1.79 \times 10^6\ \text{m/s}} = 3.13 \times 10^{-8}\ \frac{\text{kg}}{\text{C}}$$

Identify the particle from the table. All particles are completely ionized, bare nuclei.

Nucleus	m (kg)	q (C)	m/q (kg/C)
Hydrogen	1.67×10^{-27}	1.60×10^{-19}	1.04×10^{-8}
Deuterium	3.35×10^{-27}	1.60×10^{-19}	2.09×10^{-8}
Tritium	5.01×10^{-27}	1.60×10^{-19}	3.13×10^{-8}
Helium-3	5.01×10^{-27}	3.20×10^{-19}	1.57×10^{-8}

The particle is a tritium nucleus.

REMARKS The mass spectrometer is an important tool in both chemistry and physics. Unknown chemicals can be heated and ionized, and the resulting particles passed through the mass spectrometer and subsequently identified.

5.7 Magnetic Field of Long, Straight Wire And Ampère's Law

During a lecture demonstration in 1819, Danish scientist Hans Oersted (1777–1851) found that an electric current in a wire deflected a nearby compass needle. This momentous discovery, linking a magnetic field with an electric current for the first time, was the beginning of our understanding of the origin of magnetism.

A simple experiment first carried out by Oersted in 1820 clearly demonstrates that a current-carrying conductor produces a magnetic field. In this experiment, several compass needles are placed in a horizontal plane near a long vertical wire, as in Active Figure 5.20a. When there is no current in the wire, all needles point in the same direction (that of Earth's field), as one would expect. When the wire carries a strong, steady current, however, the needles all deflect in directions tangent to the circle, as in Active Figure 19.20b. These observations show that the direction of $\vec{B}$ is consistent with the following convenient rule**, right-hand rule number 2**:

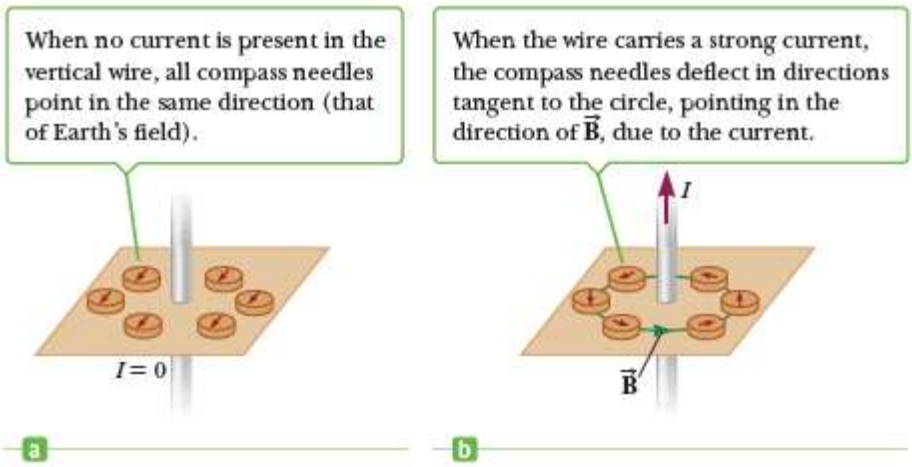


Figure 5.20

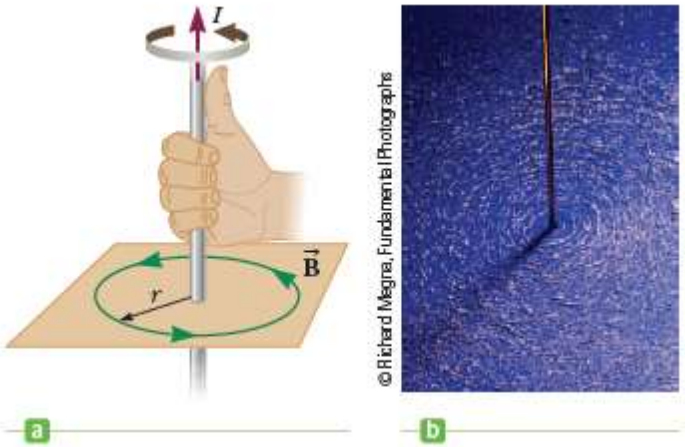


Figure 5.21

Point the thumb of your right hand along a wire in the direction of positive current, as in Figure 5.21a. Your fingers then naturally curl in the direction of the magnetic field $\vec{B}$

When the current is reversed, the filings in Figure 5.21b also reverse. Because the filings point in the direction of $\vec{B}$, we conclude that the lines of $\vec{B}$ form circles about the wire. By symmetry, the magnitude of $\vec{B}$ is the same everywhere on a circular path centered on the wire and lying in a plane perpendicular to the wire. By varying the current and distance from the wire, it can be experimentally determined that $\vec{B}$ is proportional to the current and inversely proportional to the distance from the wire. These observations lead to a mathematical expression for the strength of the magnetic field due to the current *I* in a long, straight wire:

$$B = \frac{\mu_0 I}{2\pi r} \qquad [5.11]$$

The proportionality constant μ_o, called the **permeability of free space**, has the value

$$\mu_0 \equiv 4\pi \times 10^{-7}\,\mathrm{T \cdot m/A} \qquad [5.12]$$

Ampère's Law and a Long, straight Wire

Equation 5.11 enables us to calculate the magnetic field due to a long, straight wire carrying a current. A general procedure for deriving such equations was proposed by French scientist André-Marie Ampère (1775–1836); it provides a relation between the current in an arbitrarily shaped wire and the magnetic field produced by the wire.

Consider an arbitrary closed path surrounding a current as in Figure 5.22. The path consists of many short segments, each of length Δl. Multiply one of these lengths by the component of the magnetic field parallel to that segment, where the product is labeled $B_{//}\Delta l$,. According to Ampère, the sum of all such products over the closed path is equal to μ_o times the net current I that passes through the surface bounded by the closed path. This statement, known as Ampère's circuital law, can be written

$$\sum B_{\parallel}\,\Delta\ell = \mu_0 I \qquad [5.13]$$

where $B_{//}$ is the component of $\vec{B}$ parallel to the segment of length Δl, and $\Sigma B_{//}\,\Delta l$, means that we take the sum over all the products $B_{//}\,\Delta l$, around the closed path. Ampère's law is the fundamental law describing how electric currents create magnetic fields in the surrounding empty space.

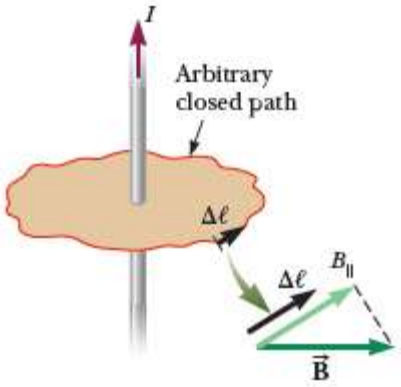


Figure 5.22

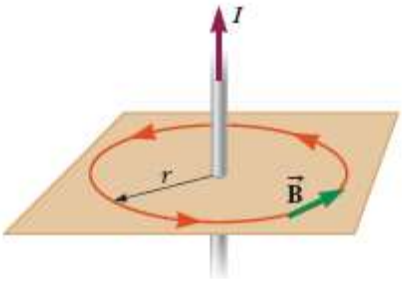


Figure 5.23

We can use Ampère's circuital law to derive the magnetic field due to a long, straight wire carrying a current I. As discussed earlier, each of the magnetic field lines of this configuration forms a circle with the wire at its centers. The magnetic field is tangent to this circle at every point, and its magnitude has the same value B over the entire circumference of a circle of radius r, so $B_{//}$ = B, as shown in Figure 5.23. In calculating the sum $\Sigma B_{//}\,\Delta l$, over the circular path, notice that $B_{//}$ can be removed from the sum (because it has the same value B for each element on the circle). Equation 5.13 then gives

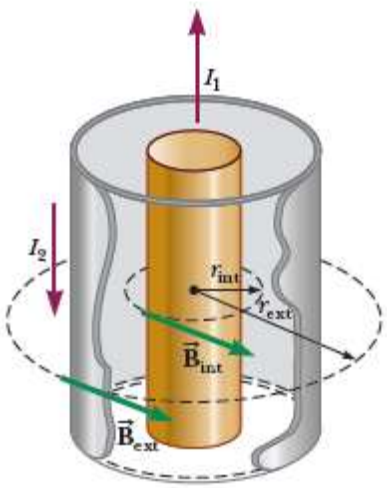
$$\sum B_{\parallel}\,\Delta\ell = B_{\parallel}\sum \Delta\ell = B(2\pi r) = \mu_0 I$$

Dividing both sides by the circumference $2\pi r$, we obtain

$$B = \frac{\mu_0 I}{2\pi r}$$

This result is identical to Equation 5.11, which is the magnetic field due to the current I in a long, straight wire. Ampère's circuital law provides an elegant and simple method for calculating the magnetic fields of highly symmetric current configurations, but it can't easily be used to calculate magnetic fields for complex current configurations that lack symmetry. In addition, Ampère's circuital law in this form is valid only when the currents and fields don't change with time.

PROBLEM A coaxial cable consists of an insulated wire carrying current I_1 = 3.00 A surrounded by a cylindrical conductor carrying current I_2 = 1.00 A in the opposite direction, as in Figure 5.24. Calculate the magnetic field inside the cylindrical conductor at r_{int} = 0.500 cm. (b) Calculate the magnetic field outside the cylindrical conductor at r_{ext} = 1.50 cm.



STRATEGY Construct a circular path around the interior wire, as in Figure 5.24. Only the current inside that circle contributes to the magnetic field B_{int} at points on the circle. To compute the magnetic field B_{ext} exterior to the cylinder, construct a circular path outside the cylinder. Now both currents must be included in the calculation, but the current going down the page must be subtracted from the current in the wire.

SOLUTION

(a) Calculate the magnetic field B_{int} inside the cylindrical conductor at $r_{int} = 0.500$ cm.

Write Ampère's law:

$$\sum B_{\parallel} \Delta \ell = \mu_0 I$$

The magnetic field is constant on the given path and the total path length is $2\pi r_{int}$:

$$B_{int}(2\pi r_{int}) = \mu_0 I_1$$

Solve for B_{int} and substitute values:

$$B_{int} = \frac{\mu_0 I_1}{2\pi r_{int}} = \frac{(4\pi \times 10^{-7}\,\mathrm{T \cdot m/A})(3.00\,\mathrm{A})}{2\pi(0.005\,\mathrm{m})}$$

$$= 1.20 \times 10^{-4}\,\mathrm{T}$$

(b) Calculate the magnetic field B_{ext} outside the cylindrical conductor at $r_{ext} = 1.50$ cm.

Write Ampère's law:

$$\sum B_{\parallel} \Delta \ell = \mu_0 I$$

The magnetic field is again constant on the given path and the total path length is $2\pi r_{ext}$:

$$B_{ext}(2\pi r_{ext}) = \mu_0 (I_1 - I_2)$$

Solve for B_{ext} and substitute values:

$$B_{ext} = \frac{\mu_0 (I_1 - I_2)}{2\pi r_{ext}} = \frac{(4\pi \times 10^{-7}\,\mathrm{T \cdot m/A})(3.00\,\mathrm{A} - 1.00\,\mathrm{A})}{2\pi(0.015\,\mathrm{m})}$$

$$= 2.67 \times 10^{-5}\,\mathrm{T}$$

REMARKS The direction of the field both inside and outside the cylinder is given by the right-hand rule, or counterclockwise in the perspective of Figure 5.24. Coaxial cables can be used to minimize the magnetic effects of current, provided that the currents inside the wire and the cylinder are equal in magnitude and opposite in direction.

5.8 Magnetic Force Between Two Parallel Conductors

As we have seen, a magnetic force acts on a current-carrying conductor when the conductor is placed in an external magnetic field. Because a conductor carrying a current creates a magnetic field around itself, it is easy to understand that two current-carrying wires placed close together exert magnetic forces on each other. Consider two long, straight, parallel wires separated by the distance d and carrying currents I_1 and I_2 in the same direction, as shown in Active Figure 5.25. Wire 1 is directly above wire 2. What's the magnetic force on one wire due to a magnetic field set up by the other wire?

In this calculation we are finding the force on wire 1 due to the magnetic field of wire 2. The current I_2 sets up magnetic field $\overrightarrow{B_2}$ at wire 1. The direction of $\overrightarrow{B_2}$ is perpendicular to the wire, as shown in the figure. Using Equation 19.11, we find that the magnitude of this magnetic field is

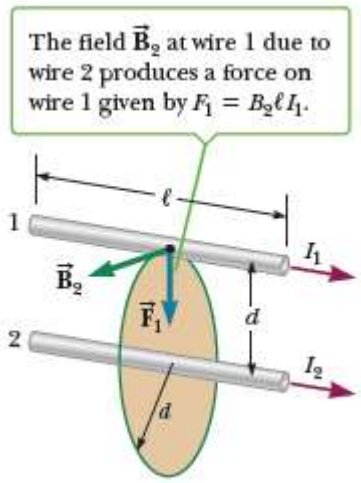
$$B_2 = \frac{\mu_0 I_2}{2\pi d}$$

According to Equation 5.5, the magnitude of the magnetic force on wire 1 in the presence of field $\vec{B_2}$ due to I_2 is

$$F_1 = B_2 I_1 \ell = \left(\frac{\mu_0 I_2}{2\pi d}\right) I_1 \ell = \frac{\mu_0 I_1 I_2 \ell}{2\pi d}$$

We can rewrite this relationship in terms of the force per unit length:

$$\frac{F_1}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d} \qquad [5.14]$$



Active Figure 5.25

The direction of $\vec{F_1}$ is downward, toward wire 2, as indicated by right-hand rule number 1. This calculation is completely symmetric, which means that the force $\vec{F_2}$ on wire 2 is equal to and opposite $\vec{F_1}$, as expected from Newton's third law of action–reaction. We have shown that parallel conductors carrying currents in the same direction attract each other. You should use the approach indicated by Figure 5.25 and the steps leading to Equation 5.14 to show that parallel conductors carrying currents in opposite directions repel each other. The force between two parallel wires carrying a current is used to define the SI unit of current, the ampere (A), as follows:

> *If two long, parallel wires 1 m apart carry the same current and the magnetic force per unit length on each wire is 2 × 10^{-7} N/m, the current is defined to be 1 A.*

The SI unit of charge, the coulomb (C), can now be defined in terms of the ampere as follows:

> *If a conductor carries a steady current of 1 A, the quantity of charge that flows through any cross section in 1 s is 1 C.*

PROBLEM Two wires, each having a weight per unit length of 1.00 × 10^{-4} N/m, are parallel with one directly above the other. Assume the wires carry currents that are equal in magnitude and opposite in direction. The wires are 0.10 m apart, and the sum of the magnetic force and gravitational force on the upper wire is zero. Find the current in the wires. (Neglect Earth's magnetic field.)

STRATEGY The upper wire must be in equilibrium under the forces of magnetic repulsion and gravity. Set the sum of the forces equal to zero and solve for the unknown current, I.

SOLUTION

Set the sum of the forces equal to zero and substitute the appropriate expressions. Notice that the magnetic force between the wires is repulsive.

$$\vec{F}_{grav} + \vec{F}_{mag} = 0$$

$$-mg + \frac{\mu_0 I_1 I_2}{2\pi d}\ell = 0$$

The currents are equal, so $I_1 = I_2 = I$. Make these substitutions and solve for I^2:

$$\frac{\mu_0 I^2}{2\pi d}\ell = mg \quad \rightarrow \quad I^2 = \frac{(2\pi d)(mg/\ell)}{\mu_0}$$

Substitute given values, finding I^2, then take the square root. Notice that the weight per unit length, mg/ℓ, is given.

$$I^2 = \frac{(2\pi \cdot 0.100\ \text{m})(1.00 \times 10^{-4}\ \text{N/m})}{(4\pi \times 10^{-7}\ \text{T} \cdot \text{m})} = 50.0\ \text{A}^2$$

$$I = 7.07\ \text{A}$$

5.9 Magnetic Fields of Current Loops and Solenoids

The strength of the magnetic field set up by a piece of wire carrying a current can be enhanced at a specific location if the wire is formed into a loop. You can understand this by considering the effect of several small segments of the current loop, as in Figure 5.26. The small segment at the bottom of the loop, labeled Δx_1, produces a magnetic field of magnitude B_1 at the loop's center, directed upward. The direction of $\vec{B}$ can be verified using right-hand rule number 2 for a long, straight wire. Imagine holding the wire with your right hand, with your thumb pointing in the direction of the current. Your fingers then curl around in the direction of $\vec{B}$

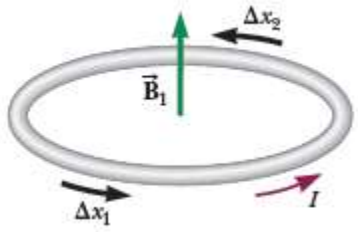


Figure 5.26

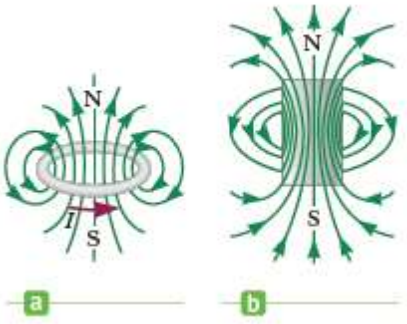


Figure 5.27

A segment of length Δx_2 at the top of the loop also contributes to the field at the center, increasing its strength. The field produced at the center by the segment Δx_2 has the same magnitude as B_1 and is also directed upward. Similarly, all other such segments of the current loop contribute to the field. The net effect is a magnetic field for the current loop as pictured in Figure 5.27a.

Notice in Figure 5.27a that the magnetic field lines enter at the bottom of the current loop and exit at the top. Compare this figure with Figure 5.27b, illustrating the field of a bar magnet. The two fields are similar. One side of the loop acts as though it were the north pole of a magnet, and the other acts as a south pole. The similarity of these two fields will be used to discuss magnetism in matter in an upcoming section.

The magnitude of the magnetic field at the center of a circular loop carrying current I is given by

$$B = \frac{\mu_0 I}{2R}$$

This equation must be derived with calculus. It can be shown, however, to be reasonable by calculating the field at the center of four long wires, each carrying current I and forming a square, as in Figure 5.28, with a circle of radius R inscribed within it. Intuitively, this arrangement should give a magnetic field at the center that is similar in magnitude to the field produced by the circular loop. The current

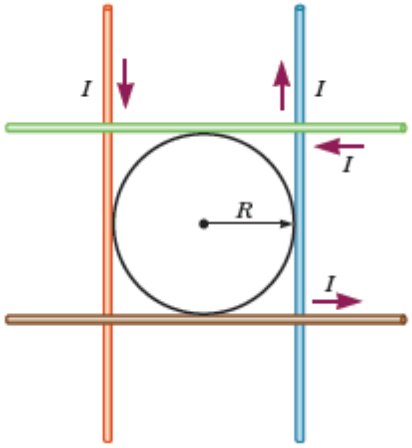


Figure 5.28

in the circular wire is closer to the center, so that wire would have a magnetic field somewhat stronger than just the four legs of the rectangle, but the lengths of the straight wires beyond the rectangle compensate for it. Each wire contributes the same magnetic field at the exact center, so the total field is given by

$$B = 4 \times \frac{\mu_0 I}{2\pi R} = \frac{4}{\pi}\left(\frac{\mu_0 I}{2R}\right) = (1.27)\left(\frac{\mu_0 I}{2R}\right)$$

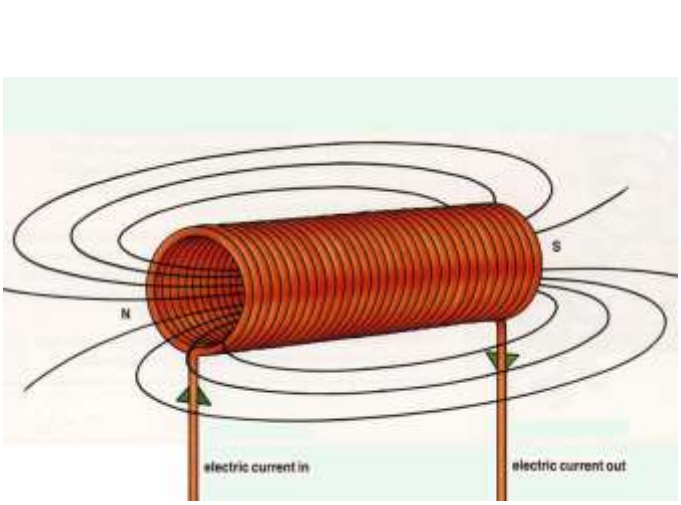
This result is approximately the same as the field produced by the circular loop of current. When the coil has N loops, each carrying current I, the magnetic field at the center is given by

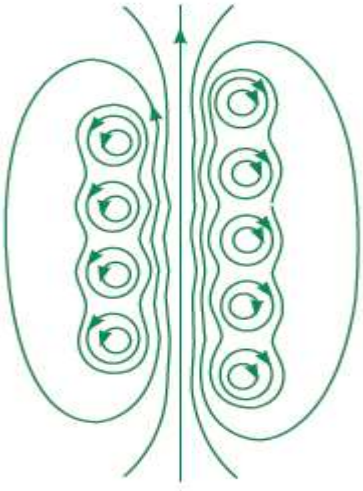
$$B = N\,\frac{\mu_0 I}{2R}$$

[5.15]

Magnetic Field of a Solenoid

If a long, straight wire is bent into a coil of several closely spaced loops, the resulting device is a **solenoid**, often called an electromagnet. This device is important in many applications because it acts as a magnet only when it carries a current. The magnetic field inside a solenoid increases with the current and is proportional to the number of coils per unit length. The following figure shows the magnetic field lines of a loosely wound solenoid of length, and total number of turns N. Notice that the field lines inside the solenoid are nearly parallel, uniformly spaced, and close together. As a result, the field inside the solenoid is strong and approximately uniform. The exterior field at the sides of the solenoid is nonuniform, much weaker than the interior field, and opposite in direction to the field inside the solenoid. If the turns are closely spaced, the field lines are as shown in the following figure entering at one end of the solenoid and emerging at the other. One end of the solenoid acts as a north pole and the other end acts as a south pole.





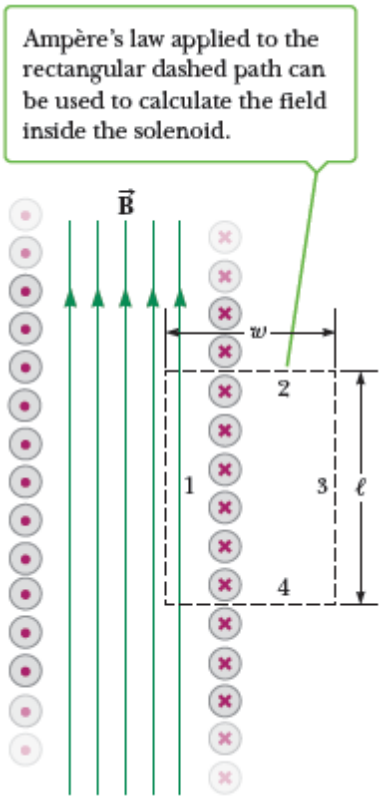
If the length of the solenoid is much greater than its radius, the lines that leave the north end of the solenoid spread out over a wide region before returning to enter the south end. The more widely separated the field lines are, the weaker the field. This is in contrast to a much stronger field inside the solenoid, where the lines are close together. Also, the field inside the solenoid has a constant

magnitude at all points far from its ends. As will be shown subsequently, these considerations allow the application of Ampère's law to the solenoid, giving a result of

$$B = \mu_0 nI \qquad [5.16]$$

for the field inside the solenoid, where n = N/l is the number of turns per unit length of the solenoid. Numerous devices create beams of charged particles for various purposes, and those particles are usually controlled and directed by electromagnetic fields. Old-style cathode ray TV sets use steering magnets that rapidly and accurately direct an electron beam across a screen of phosphors in a scanning motion, creating an illusion of a moving picture out of a series of bright dots. (See Fig. 19.34.) Electron microscopes use a similar gun and both electrostatic and electromagnetic lenses to focus the beam. Particle accelerators require very large electromagnets to turn particles moving at nearly the speed of light. Tokamaks, experimental devices used in fusion power research, use magnetic fields to contain hot plasmas.

Ampère's Law Applied to a Solenoid



We can use Ampère's law to obtain the expression for the magnetic field inside a solenoid carrying a current I. A cross section taken along the length of part of our solenoid is shown in Figure 5.31. $\vec{B}$ inside the solenoid is uniform and parallel to the axis, and $\vec{B}$ outside is approximately zero. Consider a rectangular path of length L and width w, as shown in the figure. We can apply Ampère's law to this
path by evaluating the sum of $B_{//}\ \Delta L$, over each side of the rectangle. The contribution along side 3 is clearly zero because $\vec{B}$= 0 in this region. The contributions from sides 2 and 4 are both zero because $\vec{B}$ is perpendicular to ΔL, along these paths. Side 1 of length L gives a contribution BL to the sum because $\vec{B}$ is uniform along this path and parallel to ΔL,. Therefore, the sum over the closed rectangular path has the value

$$\sum B_{\parallel}\,\Delta\ell = BL$$

The right side of Ampère's law involves the total current that passes through the area bounded by the path chosen. In this case, the total current through the rectangular path equals the current through each turn of the solenoid, multiplied by the number of turns. If N is the number of turns in the length L, then the total current through the rectangular path equals NI. Ampère's law applied to this path therefore gives

$$\sum B_{\parallel}\,\Delta\ell = BL = \mu_0 NI \qquad \text{or} \qquad B = \mu_0 \frac{N}{L} I = \mu_0 nI$$

where n = N/L is the number of turns per unit length.

5.10 Magnetic Domains

The magnetic field produced by a current in a coil of wire gives us a hint as to what might cause certain materials to exhibit strong magnetic properties. A single coil like that in Figure 5.27a has a north pole and a south pole, but if that is true for a coil of wire, it should also be true for any current confined to a circular path. In particular, *an individual atom should act as a magnet because of the motion of the electrons about the nucleus*. Each electron, with its charge of 1.6×10^{-19} C, circles the atom once in about 10^{-16} s. If we divide the electric charge by this time interval, we see that the orbiting electron is equivalent to a current of 1.6×10^{-3} A. Such a current produces a magnetic field on the order of 20 T at the center of the circular path. From this we see that a very strong magnetic field would be produced if several of these atomic magnets could be aligned inside a material. This doesn't occur, however, because the simple model we have described is not the complete story. A thorough analysis of atomic structure shows that the magnetic field produced by one electron in an atom is often canceled by an oppositely revolving electron in the same atom. The net result is that *the magnetic effect produced by the electrons orbiting the nucleus is either zero or very small for most materials.*

The magnetic properties of many materials can be explained by the fact that an electron not only circles in an orbit, but also spins on its axis like a top, with spin magnetic moment.(This classical description should not be taken too literally. The property of electron spin can be understood only in the context of quantum mechanics, which we will not discuss here.) The spinning electron represents a charge in motion that produces a magnetic field. The field due to the spinning is generally stronger than the field due to the orbital motion. In atoms containing many electrons, the electrons usually pair up with their spins opposite each other so that their fields cancel each other. That is why most substances are not magnets. In certain strongly magnetic materials, such as iron, cobalt, and nickel, however, the magnetic fields produced by the electron spins don't cancel completely. Such materials are said to be ferromagnetic. In ferromagnetic materials strong coupling occurs between neighboring atoms, forming large groups of atoms with spins that are aligned. Called domains, the sizes of these groups typically range from about 10^{-4} cm to 0.1 cm. In an unmagnetized substance the domains are randomly oriented, as shown in Figure 5.32a. When an external field is applied, as in Figures 5.32b and 5.32c, the magnetic field of each domain tends to come nearer to alignment with the external field, resulting in magnetization. In what are called hard magnetic materials, domains remain aligned even after the external field is removed; the result is a permanent magnet. In soft magnetic materials, such as iron, once the external field is removed, thermal agitation produces motion of the domains and the material quickly returns to an unmagnetized state.

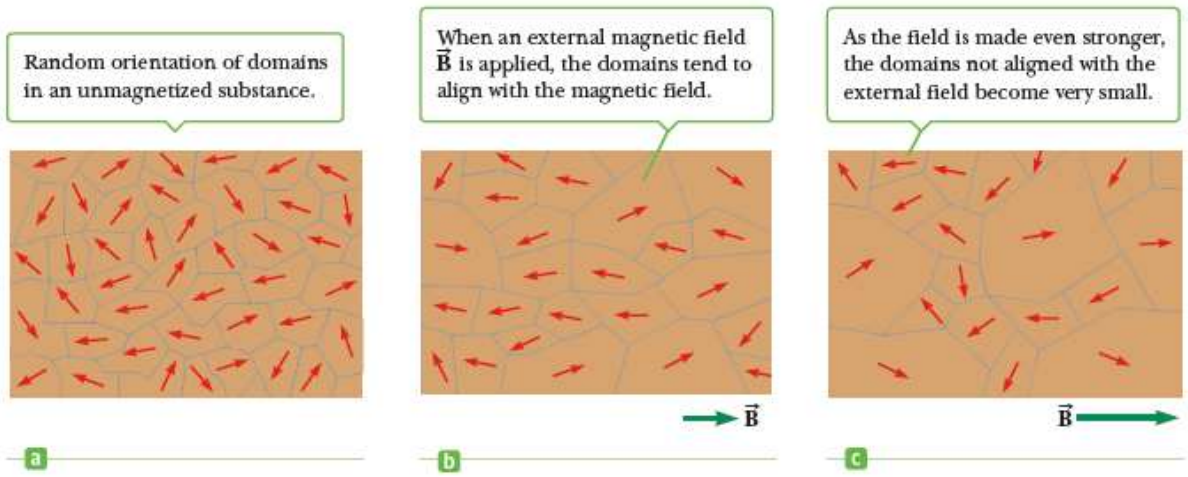


Figure 5.32

The alignment of domains explains why the strength of an electromagnet is increased dramatically by the insertion of an iron core into the magnet's center. The magnetic field produced by the current in the loops causes the domains to align, thus producing a large net external field. The use of iron as a core is also advantageous because it is a soft magnetic material that loses its magnetism almost instantaneously after the current in the coils is turned off. The formation of domains in ferromagnetic substances also explains why such substances are attracted to permanent magnets. The magnetic field of a permanent magnet realigns domains in a ferromagnetic object so that the object becomes temporarily magnetized. The object's poles are then attracted to the corresponding opposite poles of the permanent magnet. The object can similarly attract other ferromagnetic objects.

Types of Magnetic Materials

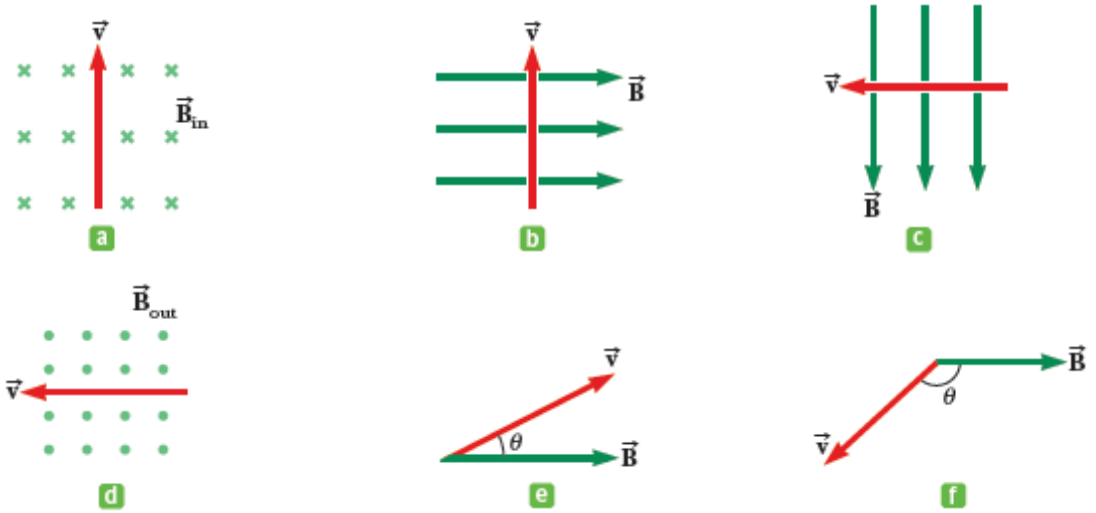
Magnetic materials can be classified according to how they react to the application of a magnetic field. In **ferromagnetic** materials the atoms have permanent magnetic moments that align readily with an externally applied magnetic field. Examples of ferromagnetic materials are iron, cobalt, and nickel. Such substances can retain some of their magnetization even after the applied magnetic field is removed.

Paramagnetic materials also have magnetic moments that tend to align with an externally applied magnetic field, but the response is extremely weak compared with that of ferromagnetic materials. Examples of paramagnetic substances are aluminum, calcium, and platinum. A ferromagnetic material can become paramagnetic when warmed to a certain critical temperature, the Curie temperature, that depends on the material.

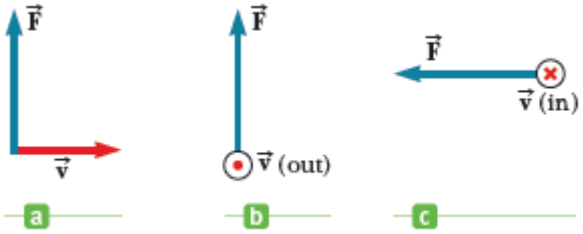
In **diamagnetic** materials, an externally applied magnetic field induces a very weak magnetization that is opposite the applied field. Ordinarily diamagnetism isn't observed because paramagnetic and ferromagnetic effects are far stronger.

PROBLEMS

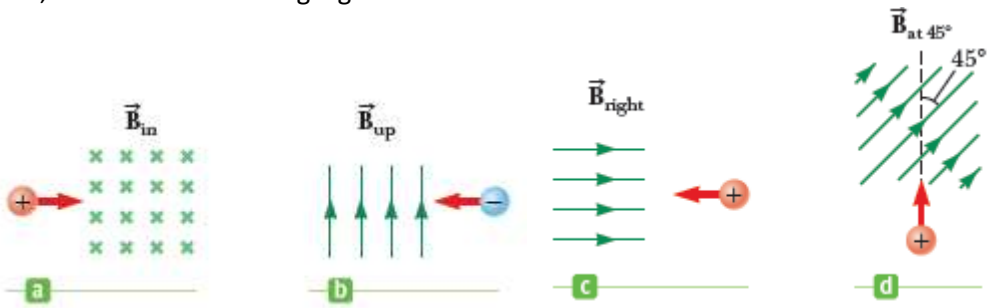
1. Consider an electron near the Earth's equator. In which direction does it tend to deflect if its velocity is (a) directed downward? (b) Directed northward? (c) Directed westward? (d) Directed southeastward?
2. (a) Find the direction of the force on a proton (a positively charged particle) moving through the magnetic fields in Figure P19.2, as shown. (b) Repeat part (a), assuming the moving particle is an electron.



3. Find the direction of the magnetic field acting on the positively charged particle moving in the various situations as shown if the direction of the magnetic force acting on it is as indicated.

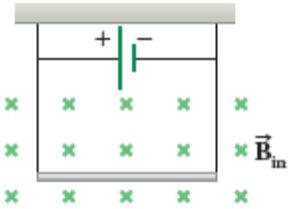


4. Determine the initial direction of the deflection of charged particles as they enter the magnetic fields, as shown in following Figure.

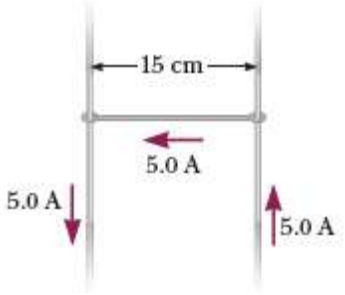


5. A laboratory electromagnet produces a magnetic field of magnitude 1.50 T. A proton moves through this field with a speed of 6.00×10^6 m/s. (a) Find the magnitude of the maximum magnetic force that could be exerted on the proton. (b) What is the magnitude of the maximum acceleration of the proton? (c) Would the field exert the same magnetic force on an electron moving through the field with the same speed? (d) Would the electron undergo the same acceleration? Explain.
6. A proton moves perpendicular to a uniform magnetic field $\vec{B}$ at a speed of 1.00×10^7 m/s and experiences an acceleration of 2.00×10^{13} m/s^2 in the positive x- direction when its velocity is in the positive z- direction. Determine the magnitude and direction of the field.
7. What velocity would a proton need to circle Earth 1000 km above the magnetic equator, where Earth's magnetic field is directed horizontally north and has a magnitude of 4.00×10^{-8} T?
8. An electron is accelerated through 2 400 V from rest and then enters a region where there is a uniform 1.70-T magnetic field. What are (a) the maximum and (b) the minimum magnitudes of the magnetic force acting on this electron?
9. A proton moving at 4.00×10^6 m/s through a magnetic field of magnitude 1.70 T experiences a magnetic force of magnitude 8.20×10^{-13} N. What is the angle between the proton's velocity and the field?
10. Sodium ions (Na^+) move at 0.851 m/s through a bloodstream in the arm of a person standing near a large magnet. The magnetic field has a strength of 0.254 T and makes an angle of 51.0° with the motion of the sodium ions. The arm contains 100 cm^3 of blood with a concentration of 3.00×10^{20} Na^+ ions per cubic centimeter. If no other ions were present in the arm, what would be the magnetic force on the arm?
11. At the equator, near the surface of Earth, the magnetic field is approximately 50.0 mT northward, and the electric field is about 100 N/C downward in fair weather. Find the gravitational, electric, and magnetic forces on an electron with an instantaneous velocity of 6.00×10^6 m/s directed to the east in this environment.
12. In the figures of problem 2 assume in each case the velocity vector shown is replaced with a wire carrying a current in the direction of the velocity vector. For each case, find the direction of the magnetic force acting on the wire.

13. A current I = 15 A is directed along the positive x-axis and perpendicular to a magnetic field. A magnetic force per unit length of 0.12 N/m acts on the conductor in the negative y-direction. Calculate the magnitude and direction of the magnetic field in the region through which the current passes.
14. A straight wire carrying a 3.0-A current is placed in a uniform magnetic field of magnitude 0.28 T directed perpendicular to the wire. (a) Find the magnitude of the magnetic force on a section of the wire having a length of 14 cm. (b) Explain why you can't determine the direction of the magnetic force from the information given in the problem.
15. In Figure of problem 3. assume in each case the velocity vector shown is replaced with a wire carrying a current in the direction of the velocity vector. For each case, find the direction of the magnetic field that will produce the magnetic force shown.
16. A wire having a mass per unit length of 0.500 g/cm carries a 2.00-A current horizontally to the south. What are the direction and magnitude of the minimum magnetic field needed to lift this wire vertically upward?
17. A wire carries a current of 10.0 A in a direction that makes an angle of 30.0° with the direction of a magnetic field of strength 0.300 T. Find the magnetic force on a 5.00-m length of the wire.
18. At a certain location, Earth has a magnetic field of 0.60 × 10-4 T, pointing 75° below the horizontal in a north–south plane. A 10.0-m-long straight wire carries a 15-A current. (a) If the current is directed horizontally toward the east, what are the magnitude and direction of the magnetic force on the wire? (b) What are the magnitude and direction of the force if the current is directed vertically upward?
19. A wire with a mass of 1.00 g/cm is placed on a horizontal surface with a coefficient of friction of 0.200. The wire carries a current of 1.50 A eastward and moves horizontally to the north. What are the magnitude and the direction of the smallest vertical magnetic field that enables the wire to move in this fashion?
20. A conductor suspended by two flexible wires as shown in following Figure has a mass per unit length of 0.040 0 kg/m. (a) What current must exist in the conductor for the tension in the supporting wires to be zero when the magnetic field is 3.60 T into the page? (b) What is the required direction for the current?



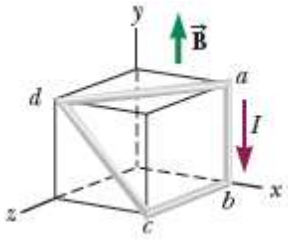
21. Consider the system pictured in following Figure. A 15-cm length of conductor of mass 15 g, free to move vertically, is placed between two thin, vertical conductors, and a uniform magnetic field acts perpendicular to the page. When a 5.0-A current is directed as shown in the figure, the horizontal wire moves upward at constant velocity in the presence of gravity. (a) What forces act on the horizontal wire, and under what condition is the wire able to move upward at constant velocity? (b) Find the magnitude and direction of the minimum magnetic field required to move the wire at constant speed. (c) What happens if the magnetic field exceeds this minimum value? (The wire slides without friction on the two vertical conductors.)



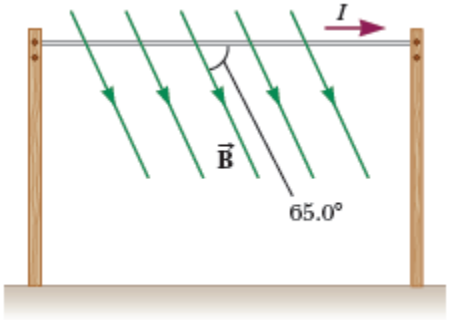
22. A metal rod of mass m carrying a current I glides on two horizontal rails a distance d apart. If the coefficient of kinetic friction between the rod and rails is μ_k, what vertical magnetic field is required to keep the rod moving at a constant speed?

23. In the following Figure the cube is 40.0 cm on each edge. Four straight segments of wire—ab, bc, cd, and da—form a closed loop that carries a current I = 5.00 A in the direction shown. A

uniform magnetic field of magnitude B = 0.020 0 T is in the positive y-direction. Determine the magnitude and direction of the magnetic force on each segment.



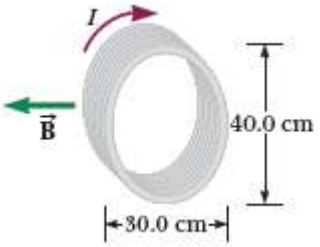
24. A horizontal power line of length 58 m carries a current of 2.2 kA as shown in the following Figure. Earth's magnetic field at this location has a magnitude equal to 5.0×10^{-5} T and makes an angle of 65° with the power line. Find the magnitude and direction of the magnetic force on the power line.



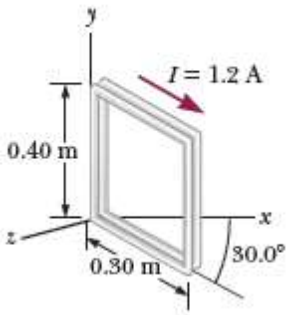
25. A wire is formed into a circle having a diameter of 10.0 cm and is placed in a uniform magnetic field of 3.00 mT. The wire carries a current of 5.00 A. Find the maximum torque on the wire.

26. A current of 17.0 mA is maintained in a single circular loop with a circumference of 2.00 m. A magnetic field of 0.800 T is directed parallel to the plane of the loop. What is the magnitude of the torque exerted by the magnetic field on the loop?

27. An eight-turn coil encloses an elliptical area having a major axis of 40.0cm and a minor axis of 30.0 cm as the following figure. The coil lies in the plane of the page and carries a clockwise current of 6.00 A. If the coil is in a uniform magnetic field of 2.00 3 1024 T directed toward the left of the page, what is the magnitude of the torque on the coil? Hint: The area of an ellipse is A =πab, where a and b are, respectively, the semimajor and semiminor axes of the ellipse.

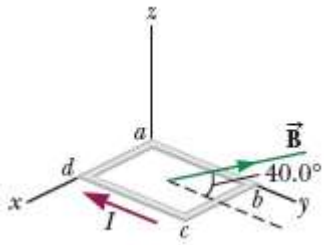


28. A rectangular loop consists of 100 closely wrapped turns and has dimensions 0.400 m by 0.300 m. The loop is hinged along the y-axis, and the plane of the coil makes an angle of 30.0° with the x-axis. What is the magnitude of the torque exerted on the loop by a uniform magnetic field of 0.800 T directed along the x-axis when the current in the windings has a value of 1.20 A in the direction shown? What is the expected direction of rotation of the loop?

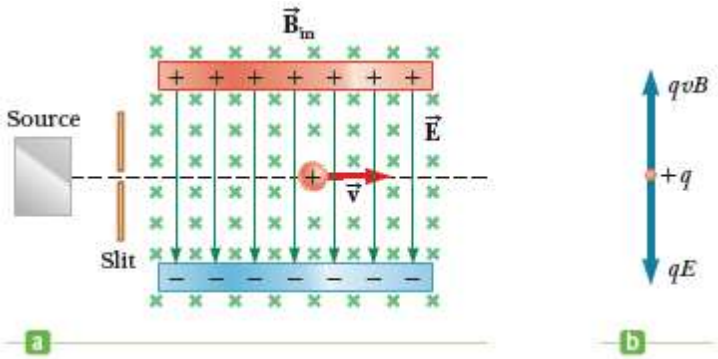


29. A 200-turn rectangular coil having dimensions of 3.0 cm by 5.0 cm is placed in a uniform magnetic field of magnitude 0.90 T. (a) Find the current in the coil if the maximum torque exerted on it by the magnetic field is 0.15 N ? m. (b) Find the magnitude of the torque on the coil when the magnetic field makes an angle of 25° with the normal to the plane of the coil.

30. A copper wire is 8.00 m long and has a cross-sectional area of 1.00×10^{-4} m^2. The wire forms a one-turn loop in the shape of square and is then connected to a battery that applies a potential difference of 0.100 V. If the loop is placed in a uniform magnetic field of magnitude 0.400 T, what is the maximum torque that can act on it? The resistivity of copper is $1.70 \times 10^{-8}\ \Omega$ m.
31. A long piece of wire with a mass of 0.100 kg and a total length of 4.00 m is used to make a square coil with a side of 0.100 m. The coil is hinged along a horizontal side, carries a 3.40-A current, and is placed in a vertical magnetic field with a magnitude of 0.010 0 T. (a) Determine the angle that the plane of the coil makes with the vertical when the coil is in equilibrium. (b) Find the torque acting on the coil due to the magnetic force at equilibrium.
32. A rectangular loop has dimensions 0.500 m by 0.300 m. The loop is hinged along the x-axis and lies in the xy-plane. A uniform magnetic field of 1.50 T is directed at an angle of 40.0° with respect to the positive y-axis and lies parallel everywhere to the yz-plane. The loop carries a current of 0.900 A in the direction shown. (Ignore gravitation.) (a) In what direction is magnetic force exerted on wire segment ab? What is the direction of the magnetic torque associated with this force, as computed with respect to the x-axis? (b) What is the direction of the magnetic force exerted on segment cd? What is the direction of the magnetic torque associated with this force, again computed with respect to the x-axis? (c) Can the forces examined in parts (a) and (b) combine to cause the loop to rotate around the x-axis? Can they affect the motion of the loop in any way? Explain. (d) What is the direction (in the yz-plane) of the magnetic force exerted on segment bc? Measuring torques with respect to the x-axis, what is the direction of the torque exerted by the force on segment bc? (e) Looking toward the origin along the positive x-axis, will the loop rotate clockwise or counterclockwise? (f) Compute the magnitude of the magnetic moment of the loop. (g) What is the angle between the magnetic moment vector and the magnetic field? (h) Compute the torque on the loop using the values found for the magnetic moment and magnetic field.

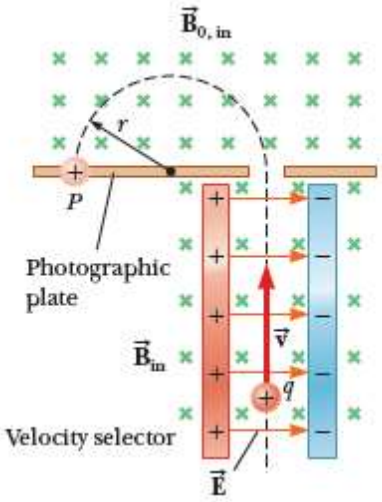


33. An electron moves in a circular path perpendicular to a magnetic field of magnitude 0.235 T. If the kinetic energy of the electron is 3.30×10^{-19} J, find (a) the speed of the electron and (b) the radius of the circular path.
34. A proton travels with a speed of 5.02×10^6 m/s at an angle of 60° with the direction of a magnetic field of magnitude 0.180 T in the positive x-direction. What are (a) the magnitude of the magnetic force on the proton and (b) the proton's acceleration?
35. Figure as follow is a diagram of a device called a velocity selector, in which particles of a specific velocity pass through undeflected while those with greater or lesser velocities are deflected either upwards or downwards. An electric field is directed perpendicular to a magnetic field, producing an electric force and a magnetic force on the charged particle that can be equal in magnitude and opposite in direction and hence cancel. Show that particles with a speed of v = E/B will pass through the velocity selector undeflected.

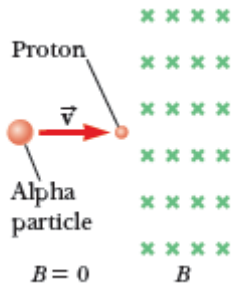


36. Consider the mass spectrometer shown schematically in the following Figure. The electric field between the plates of the velocity selector is 950 V/m, and the magnetic fields in both the velocity selector and the deflection chamber have magnitudes of 0.930 T. Calculate the radius

of the path in the system for a singly charged ion with mass m = 2.18 × 10^{-26} kg. Hint: See Problem 35.

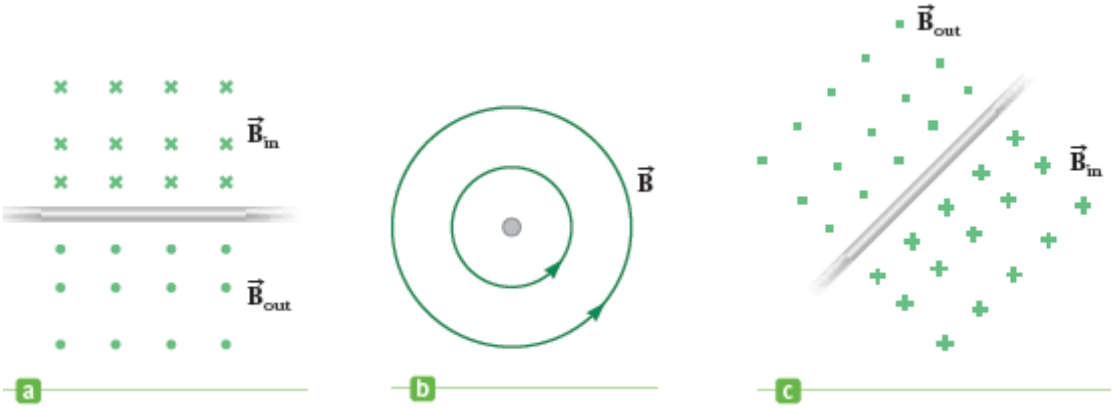


37. A singly charged positive ion has a mass of 2.50 ×10^{-26} kg. After being accelerated through a potential difference of 250 V, the ion enters a magnetic field of 0.500 T, in a direction perpendicular to the field. Calculate the radius of the path of the ion in the field.
38. A mass spectrometer is used to examine the isotopes of uranium. Ions in the beam emerge from the velocity selector at a speed of 3.00 × 10^5 m/s and enter a uniform magnetic field of 0.600 T directed perpendicularly to the velocity of the ions. What is the distance between the impact points formed on the photographic plate by singly charged ions of ^{235}U and ^{238}U?
39. A proton is at rest at the plane vertical boundary of a region containing a uniform vertical magnetic field B. An alpha particle moving horizontally makes a head-on elastic collision with the proton. Immediately after the collision, both particles enter the magnetic field, moving perpendicular to the direction of the field. The radius of the proton's trajectory is R. The mass of the alpha particle is four times that of the proton, and its charge is twice that of the proton. Find the radius of the alpha particle's trajectory.

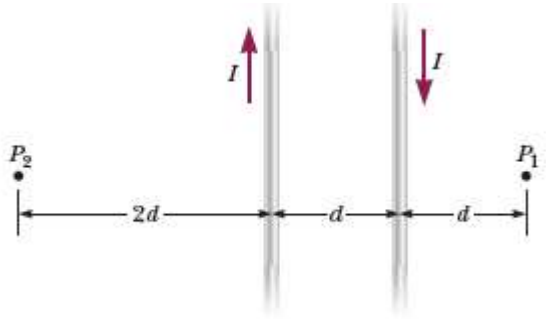


40. A particle with charge q and kinetic energy KE travels in a uniform magnetic field of magnitude B. If the particle moves in a circular path of radius R, find expressions for (a) its speed and (b) its mass.
41. A particle passes through a mass spectrometer as illustrated in Figure of problem 36. The electric field between the plates of the velocity selector has a magnitude of 8 250 V/m, and the magnetic fields in both the velocity selector and the deflection chamber have magnitudes of 0.093 1 T. In the deflection chamber the particle strikes a photographic plate 39.6 cm removed from its exit point after traveling in a semicircle. (a) What is the mass-to-charge ratio of the particle? (b) What is the mass of the particle if it is doubly ionized? (c) What is its identity, assuming it's an element?

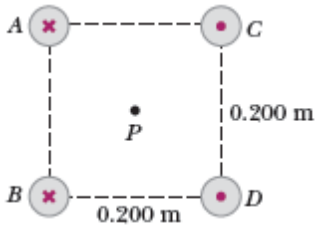
42. A proton (charge +e, mass m_p), a deuteron (charge +e, mass $2m_p$), and an alpha particle (charge +2e, mass $4m_p$) are accelerated from rest through a common potential difference ΔV. Each of the particles enters a uniform magnetic field $\vec{B}$, with its velocity in a direction perpendicular to $\vec{B}$. The proton moves in a circular path of radius r_p. In terms of r_p, determine (a) the radius rd of the circular orbit for the deuteron and (b) the radius r_a for the alpha particle.
43. A lightning bolt may carry a current of 1.00×10^4 A for a short time. What is the resulting magnetic field 100 m from the bolt? Suppose the bolt extends far above and below the point of observation.
44. In each of parts (a), (b), and (c) of the following Figure, find the direction of the current in the wire that would produce a magnetic field directed as shown.



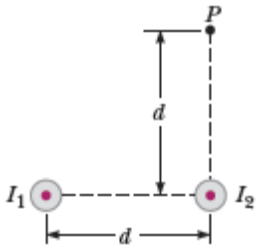
45. Neurons in our bodies carry weak currents that produce detectable magnetic fields. A technique called magnetoencephalography, or MEG, is used to study electrical activity in the brain using this concept. This technique is capable of detecting magnetic fields as weak as 1.0×10^{-15} T. Model the neuron as a long wire carrying a current and find the current it must carry to produce a field of this magnitude at a distance of 4.0 cm from the neuron.
46. In 1962 measurements of the magnetic field of a large tornado were made at the Geophysical Observatory in Tulsa, Oklahoma. If the magnitude of the tornado's field was 1.50×10^{-8} T pointing north when the tornado was 9.00 km east of the observatory, what current was carried up or down the funnel of the tornado? Model the vortex as a long, straight wire carrying a current.
47. A cardiac pacemaker can be affected by a static magnetic field as small as 1.7 mT. How close can a pacemaker wearer come to a long, straight wire carrying 20 A?
48. The two wires shown in following Figure are separated by d = 10.0 cm and carry currents of I = 5.00 A in opposite directions. Find the magnitude and direction of the net magnetic field (a) at a point midway between the wires; (b) at point P_1, 10.0 cm to the right of the wire on the right; and (c) at point P_2, 2d = 20.0 cm to the left of the wire on the left.



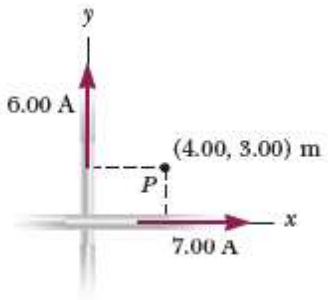
49. Four long, parallel conductors carry equal currents of I = 5.00 A. The following Figure is an end view of the conductors. The direction of the current is into the page at points A and B (indicated by the crosses) and out of the page at C and D (indicated by the dots). Calculate the magnitude and direction of the magnetic field at point P, located at the center of the square with edge of length 0.200 m.



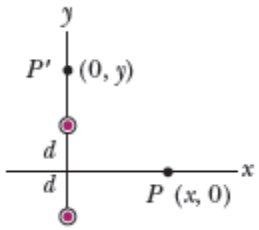
50. Two long, parallel wires carry currents of I_1 = 3.00 A and I_2 = 5.00 A in the direction indicated in the following Figure. (a) Find the magnitude and direction of the magnetic field at a point midway between the wires (d = 20.0 cm). (b) Find the magnitude and direction of the magnetic field at point P, located d = 20.0 cm above the wire carrying the 5.00-A current.



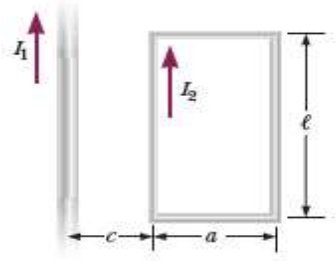
51. A wire carries a 7.00-A current along the x-axis, and another wire carries a 6.00-A current along the y-axis, as shown in following Figure. What is the magnetic field at point P, located at x = 4.00 m, y = 3.00 m?



52. A long, straight wire lies on a horizontal table in the xy-plane and carries a current of 1.20 mA in the positive x-direction along the x-axis. A proton is traveling in the negative x-direction at speed 2.30×10^4 m/s a distance d above the wire (i.e., z = d). (a) What is the direction of the magnetic field of the wire at the position of the proton? (b) What is the direction of the magnetic force acting on the proton? (c) Explain why the direction of the proton's motion doesn't change. (d) Using Newton's second law, find a symbolic expression for d in terms of the acceleration of gravity g, the proton mass m, its speed v, charge q, and the current I. (e) Find the numeric answer for the distance d using the results of part (d).
53. The magnetic field 40.0 cm away from a long, straight wire carrying current 2.00 A is 1.00 mT. (a) At what distance is it 0.100mT? (b) At one instant, the two conductors in a long household extension cord carry equal 2.00-A currents in opposite directions. The two wires are 3.00 mm apart. Find the magnetic field 40.0 cm away from the middle of the straight cord, in the plane of the two wires. (c) At what distance is it one-tenth as large? (d) The center wire in a coaxial cable carries current 2.00 A in one direction, and the sheath around it carries current 2.00 A in the opposite direction. What magnetic field does the cable create at points outside?
54. Two long, parallel wires separated by a distance 2d carry equal currents in the same direction. An end view of the two wires is shown in following Figure, where the currents are out of the page. (a) What is the direction of the magnetic field at P on the x-axis set up by the two wires? (b) Find an expression for the magnitude of the field at P. (c) From your result to part (b), determine the field at a point midway between the two wires. Does your result meet with your expectation? Explain.



55. Two long, parallel wires separated by 2.50 cm carry currents in opposite directions. The current in one wire is 1.25 A, and the current in the other is 3.50 A. (a) Find the magnitude of the force per unit length that one wire exerts on the other. (b) Is the force attractive or repulsive?
56. Two parallel wires separated by 4.0 cm repel each other with a force per unit length of 2.0×10^{-4} N/m. The current in one wire is 5.0 A. (a) Find the current in the other wire. (b) Are the currents in the same direction or in opposite directions? (c) What would happen if the direction of one current were reversed and doubled?
57. A wire with a weight per unit length of 0.080 N/m is suspended directly above a second wire. The top wire carries a current of 30.0 A, and the bottom wire carries a current of 60.0 A. Find the distance of separation between the wires so that the top wire will be held in place by magnetic repulsion.
58. In the following Figure the current in the long, straight wire is I_1 = 5.00 A, and the wire lies in the plane of the rectangular loop, which carries 10.0 A. The dimensions shown are c = 0.100 m, a = 0.150 m, and l = 0.450 m. Find the magnitude and direction of the net force exerted by the magnetic field due to the straight wire on the loop.



59. A long solenoid that has 1 000 turns uniformly distributed over a length of 0.400 m produces a magnetic field of magnitude 1.00×10^{-4} T at its center. What current is required in the windings for that to occur?
60. A certain superconducting magnet in the form of a solenoid of length 0.50 m can generate a magnetic field of 9.0 T in its core when its coils carry a current of 75 A. The windings, made of a niobium–titanium alloy, must be cooled to 4.2 K. Find the number of turns in the solenoid.
61. It is desired to construct a solenoid that will have a resistance of 5.00 V (at 20°C) and produce a magnetic field of 4.00×10^{-2} T at its center when it carries a current of 4.00 A. The solenoid is to be constructed from copper wire having a diameter of 0.500 mm. If the radius of the solenoid is to be 1.00 cm, determine (a) the number of turns of wire needed and (b) the length the solenoid should have.
62. A solenoid 10.0 cm in diameter and 75.0 cm long is made from copper wire of diameter 0.100 cm, with very thin insulation. The wire is wound onto a cardboard tube in a single layer, with adjacent turns touching each other. What power must be delivered to the solenoid if it is to produce a field of 8.00 mT at its center?
63. An electron is moving at a speed of 1.0×10^4 m/s in a circular path of radius 2.0 cm inside a solenoid. The magnetic field of the solenoid is perpendicular to the plane of the electron's path. Find (a) the strength of the magnetic field inside the solenoid and (b) the current in the solenoid if it has 25 turns per centimeter.